Eagle-Eyed Multitask CNNs for Aerial Image Retrieval and Scene Classification

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***Abstract*— In recent years, convolutional neural networks (CNNs) have become the predominant method for content-based aerial image retrieval (CBAIR) and aerial scene classification (ASC) due to their overwhelming performance advantages. However, existing CNN-based models have the following shortcomings: first, they do not deal with large intraclass variations, thereby overlooking the possibility of fine-grained retrieval and classification; second, all similarity learning methods for CBAIR consider similarity between two images as a constant, neglecting the fact that image similarity is uncertain in nature; third, similarity learning is separated from ASC, ignoring the advantages of joint optimization. To address these issues, we propose a novel metric learning method called center-metric learning, and couple it with a new kind of loss called positive-negative center loss, which, with the help of several "experts," enables CNNs to cope successfully with within-class variations. Besides, we propose similarity distribution learning, making the first attempt to embed uncertainty regarding similarity into the training process. The resulting fine-grained similarity predictions can further strengthen CNNs' fine discrimination ability. Furthermore, three tasks, that is, center-metric learning, similarity distribution learning, and ASC, are incorporated into one CNN, benefitting from one another and leading to a better generalization capability. Just like an eagle, our model is able to discriminate subtle differences among aerial images, hence the name "eagle-eyed multitask CNN." We carry out extensive experiments over four publicly available aerial image sets and achieve a performance better than all existing methods.**

***Index Terms*— Center-metric learning, eagle-eyed multitask convolutional neural network (EM-CNN), fine-grained retrieval, positive–negative center loss, similarity distribution learning.**

I. Introduction

THE recent advances in sensor technology have resulted in tremendous growth of aerial image archives. This has motivated increasing research interest in content-based aerial image retrieval (CBAIR) and aerial scene classification (ASC). CBAIR, which aims to find visually similar images for a given query from a very large aerial image archive [1], plays a crucial role in answering the urgent need for intelligent aerial data management. ASC, which aims to automatically label an aerial image according to a given set of semantic categories [2], is a fundamental task for automatic aerial image understanding and can find a wide range of applications, for instance, in urban planning, land resource management, and computer cartography.

This article develops a framework in which CBAIR and ASC are simultaneously dealt with.

A. State of the Art

In both CBAIR and ASC, the most important step is to generate discriminative and descriptive features. Features mainly consist of traditional ones and deep ones. Traditional features are hand-engineered, whereas deep features are automatically learned from data by deep learning models, of which convolutional neural network (CNN) is most representative. In recent years, CNN-based methods have overwhelmingly predominated over the traditional ones due to their unprecedented performance. This article focuses on CNNs; as for the traditional methods, we refer readers to [2], [3] for a comprehensive survey and to [4]–[7] for the latest research.

CNN feature generation approaches for ASC can be broadly divided into five groups.

1) Treating the CNNs pretrained over the everyday image set ImageNet [8] as ready-made feature extractors and directly using the activations of fully connected (FC) layers [3], [9]–[11].
2) Fine-tuning the pretrained CNNs using aerial images [3], [11].
3) Extracting information from the pretrained CNNs and then reprocessing it. For example, Yuan *et al.* [12] extracted local features from Visual Geometry Group (VGG)-VD-19 [13] and rearranged them by aggregating aerial images; He *et al.* [14] proposed "multilayer stacked covariance pooling" and used the resulting covariance matrices as final features. In most cases, the traditional methods were leveraged to fully exploit the potential of deep features. For instance, Zhu *et al.* [15] integrated fully sparse topic models with CNNs; Flores *et al.* [16] utilized a Gaussian mixture model to generate robust and compact "dictionaries of deep features;" Yu and Liu [17] and Anwer *et al.* [18] combined traditional hand-crafted features with deep features; besides, some researchers [9], [19], [20] employed feature encoding methods, such as bag of visual words (BoVWs), vector of locally aggregated descriptors (VLADs), and/or improved Fisher kernel (IFK), to transform the dense

Manuscript received July 15, 2019; revised January 11, 2020 and February 25, 2020; accepted March 2, 2020. Date of publication March 18, 2020; date of current version August 28, 2020. This work was supported in part by NSFC under Grant 61673184 and in part by the China Scholarship Council under Grant 201806755003. *(Corresponding author: Yishu Liu.)*

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Digital Object Identifier 10.1109/TGRS.2020.2979011

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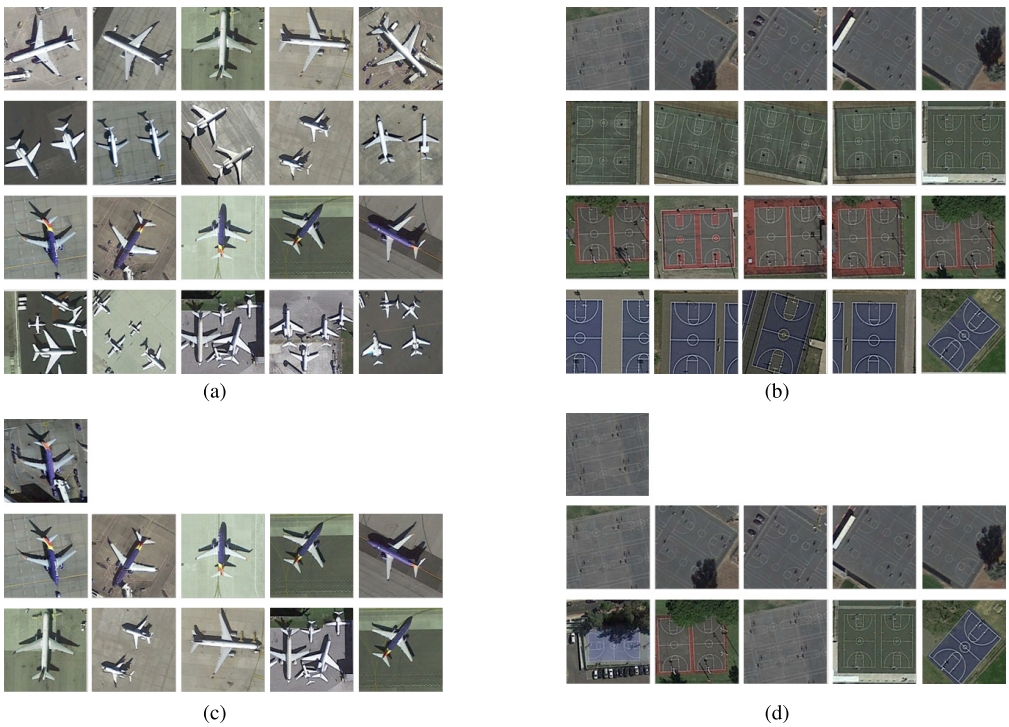


Fig. 1. Large intraclass variations and fine- and coarse-grained retrieval. (a) Some images from the "airplane" class in PatternNet. Images in the same group/row are visually alike, whereas those from different groups look very different. (b) Some images from the "basketball court" class in PatternNet. (c) "Airplane" query and its fine- and coarse-grained ranking lists. All retrieved images come from the same class. Images in the fine-grained ranking list belong to the same group, whereas those in the coarse-grained ranking list belong to different groups. Existing CBAIR methods treat fine- and coarse-grained retrieval results indiscriminately, considering them equally desirable. However, fine-grained retrieval results are apparently more satisfying than coarse-grained ones. (d) "Basketball court" query and its fine- and coarse-grained ranking lists.

convolutional activations into holistic feature vectors (FVs). Additionally, a variety of fusion strategies were proposed [9], [10], [21], [22].

4) Converting and retraining the existing CNNs. Some researchers refitted the pretrained CNNs and then retrained them. For example, Xie *et al.* [23] proposed a CNN capable of classifying aerial images of arbitrary size by converting FC layers into convolutional layers, which were connected to a global average pooling layer; inspired by the human visual system, Wang *et al.* [24] transformed the pretrained CNNs into the so-called "attention recurrent convolutional networks"; Liu *et al.* [25] consolidated information from intermediate and FC layers to create a novel CNN with directed acyclic graph topology. Moreover, some effective strategies such as contrastive embedding [26], deep structural metric learning [27], semisupervised generative frameworks [28], and Wasserstein distance [29], [30] were used to regularize the training procedure, enforcing CNNs to learn more discriminative features.
5) Building new CNNs and training them from scratch [2], [31]–[33].

CNN-based methods for CBAIR are basically the same as those for ASC, falling into the five aforementioned groups, whose examples are as follows: [34]–[36] for group 1, [37]–[39] for group 2, [40]–[42] for group 3, [43]–[45] for group 4, and [1], [46], [47] for group 5.

B. Drawbacks of Existing Methods

Section I-A summarizes the existing CNN-based approaches for ASC and CBAIR. In our opinion, these methods have the following shortcomings.

1) They all ignore intraclass variations. Owing to extraordinarily complex geometrical structures and spatial patterns, aerial images always have large intraclass visual variations (see Fig. 1 for some examples). Fig. 1(a) presents 20 images from the "airplane" class in PatternNet (see Section III for more details of the aerial image set PatternNet). Obviously, images in the same row/group seem quite similar, whereas those from different groups look markedly different. The same goes for Fig. 1(b), which shows 20 images from the "basketball court" class in PatternNet. We can see that aerial images belonging to the same class may or may not be "close together," and they can be disaggregated into several groups so that there is a big disparity across groups. Apparently, in Fig. 1(c) and (d), with the images in the first rows serving as queries, the retrieval results in the second rows are more satisfactory than those in the third rows. However, the existing CBAIR methods pay no attention to intraclass variations, that is, disparities across groups, they indiscriminately deal with these different retrieval results, considering them equally desirable.

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Similarly, all existing ASC methods only aim for between-class separability and completely disregard within-class variations. Undoubtedly, an ASC model can learn more discriminative and powerful features, if it is forced to distinguish among different groups (hereafter, we call groups subclasses) within one class.

Can we develop a CNN model capable of "fine-grained ASC" and "fine-grained CBAIR"? Can we endow a CNN model with eyesight as keen as that of an eagle?

2) Similarity learning is separated from ASC. CBAIR generally includes two main steps: feature generation and similarity measurement (at the second stage, the generated features are used to compute similarities between a query and the aerial images in the archive and retrieve the most relevant images). Most of the existing CNN-based CBAIR models, or more precisely, the first four groups of methods, are classification-oriented, that is, tasks at different stages are carried out separately, and the CNNs used for feature learning were originally meant for image classification rather than image retrieval, excluding similarity measurement from the training process. This poses a difficulty in generating tailored features that can faithfully embody what users consider "similar." After all, features fit for classification are not necessarily congenial to retrieval tasks.

   The fifth group of CBAIR models [46], [47] are an exception—they can simultaneously learn expressive features and find an appropriate similarity measurement, with an optimization objective constructed under similarity constraints. Therefore, this group of methods, developed in the setting of image retrieval, are more natural and suitable for CBAIR. However, since they do not aim for ASC, semantic labels can not be taken full advantage of. In a way, this will impair CNNs' generalization ability.

   Can we develop a CNN model in which CBAIR (more precisely, similarity learning) and ASC interact with and benefit from each other?

3) All the existing similarity learning models ignore uncertainty as to image similarity. As mentioned above, only the fifth group of CBAIR approaches were developed for the purpose of learning similarity. However, the models in [46] and [47] consider image similarity to be two-valued ("similar" or "dissimilar") and, correspondingly, can yield only binary predictions. Undoubtedly, to predict how similar two images are is more difficult and desirable than to predict whether two images are similar.

   Fortunately, Liu *et al.* [1] proposed a model called similarity-based Siamese (SBS)-CNN, which can yield predictions ranging continuously from 0 to 1, quantitatively describing the extent to which two images are alike. This is encouraging because "continuous predictions" bring us one step closer to fine-grained retrieval.

   However, SBS-CNN regards ground truth similarity as a constant, trying to narrow the gap between ground truth and prediction during training. We argue that the similarity between two images involves uncertainty (i.e., it has many possible values) and, hence, it may be inappropriate to treat it as a fixed value. As an example, consider the three images in Fig. 2, which come from the "medium residential" class. We asked 50 people the following question: which is more similar to image 1—image 2 or image 3? Then, 31 people answered "image 3," 13 people answered "image 2," and the remaining six people answered "either." This indicates that human beings have divergent views on image similarity. Similarly, the experimental results in [1] demonstrate that for a given image pair, similarity scores (SSs) computed by different algorithms/models are always unequal. Considering the divergence of opinion among human beings and among algorithms, treating image similarity as a constant value is not a good way to learn similarity.

   Can we develop a CNN model capable of both yielding continuous predictions and modeling similarity uncertainty?



(1) (2) (3)

Fig. 2. Uncertainty about similarity. When asked "which is more similar to image 1—image 2 or image 3?" 31 people replied "image 3," 13 people replied "image 2," and the remaining six people replied "either." This indicates that there exists uncertainty regarding similarity between two images.

We address all these issues in this article.

C. Related Work

1) Center Loss and Metric Learning: All the CNN models in this article use stochastic gradient descent (SGD) [48] with mini-batch updates as the optimization algorithm. Suppose that there are N classes, and denote by x and $Y(x) \in \{1, 2, \ldots, N\}$ an aerial image and its class label, respectively.

In ASC, we can consider the outputs of a CNN's softmax layer as an N-dimensional vector function

$$p(x) = (p_1(x), p_2(x), \ldots, p_N(x))^{\mathbf{T}} \tag{1}$$

where $p_j(x)$ denotes the predicted probability that x belongs to the jth class ($j \in \{1, 2, \ldots, N\}$) and $\mathbf{T}$ means transpose. Then, for a mini-batch of aerial images, $\mathscr{B}$, the most commonly used cross-entropy (CE) loss is defined as

$$\mathcal{L}_{\text{ce}}(\omega; \mathscr{B}) = -\sum_{x \in \mathscr{B}} \log p_{Y(x)}(x) \tag{2}$$

where ω denotes CNN parameters (including weights and biases). Let

$$\mathcal{E}(x) = -\log p_{Y(x)}(x) \tag{3}$$

then CE loss can also be written as

$$\mathcal{L}_{\text{ce}}(\omega; \mathscr{B}) = \sum_{x \in \mathscr{B}} \mathcal{E}(x). \tag{4}$$

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From (2), we can see that CE loss focuses on learning features that can separate one class from others, not caring whether the features within one class are widely scattered [see Fig. 3(a)].

Center loss [49] compensates for CE loss' weaknesses by simultaneously learning a center for each class and penalizing the distances between features and their corresponding centers. In this way, features for the same class are gathered tightly around the class center and, hence, leading to greater discrimination [see Fig. 3(b) for an intuitive explanation]. We consider a CNN as a function f that maps every image x to a high-dimensional FV $f(x)$ [strictly speaking, $f(x)$ should be written as $f(\omega; x)$; to simplify the notation, we drop "ω"], then center loss can be defined as

$$\mathcal{L}_{\text{center}}(\omega; \mathscr{B}) = \frac{1}{2} \sum_{x \in \mathscr{B}} \left\| f(x) - \Theta_{Y(x)} \right\|_2^2 \tag{5}$$

where $\Theta_{Y(x)}$ denotes the $Y(x)$th class center, which, in each iteration, is computed by averaging the features corresponding to the $Y(x)$th class.

Despite increasing within-class closeness, center loss does not pay attention to between-class separability. In contrast, metric learning-based loss, such as contrastive loss [26], [50] and triplet loss [2], [51], aims for both within-class pulling and across-class pushing [see Fig. 3(c)]. However, this kind of loss has the following deficiencies: 1) it needs image pairs/triplets as inputs, while image pair/triplet generation, which determines the system's performance to some extent, is very tricky and remains an open question, 2) a huge number of image pairs/triplets are needed for training, consuming much more training time than when center loss is used, and 3) hard sample mining is usually necessary for a desirable performance; unfortunately, just like image pair/triplet generation, hard sample mining is a prickly problem.

In this article, we propose center-metric learning and couple it with a new kind of loss called positive–negative center loss, which is constructed on the basis of center loss and triplet loss, not only inheriting their advantages but also remedying their deficiencies. Positive–negative center loss includes level-1 positive–negative center (L1PNC) loss, level-2 positive–negative center (L2PNC) loss, and multilevel positive–negative center (MLPNC) loss. L2PNC loss is a fine-grained version of L1PNC loss, aiming to cope with intraclass variations and distinguish among different subclasses. MLPNC loss, an integration of L1PNC and L2PNC losses, can endow CNNs with subtle discernment (see Fig. 3(d) for an intuitive understanding).

2) Similarity Learning and Age Distribution Learning: In [1], with several widely used CNNs (called experts) providing ground truth similarities, Siamese networks can be trained in the setting of deep regression [52], [53], aiming to predict similarities for image pairs. Since deep regression is sensitive to outliers and prone to unstable predictions, Liu *et al.* [1] transformed deep regression into deep ordinal classification and achieved a state-of-the-art retrieval performance. However, both similarity regression and deep ordinal classification consider image similarity a constant, which is unreasonable, as discussed in Section I-B.

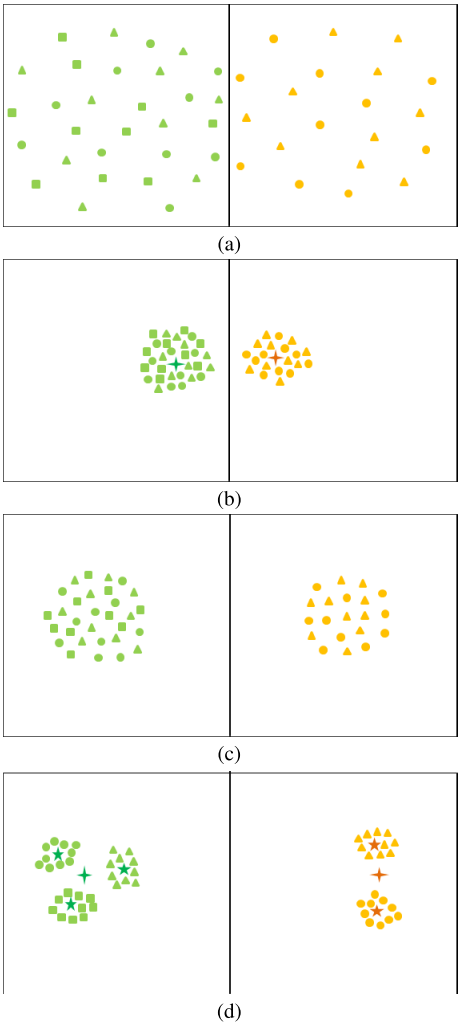


Fig. 3. Aims of different losses. (a) CE loss. CE loss concentrates only on class separability, not caring about intraclass compactness. (b) Center loss. Center loss aims for intraclass compactness by simultaneously learning a center for each class and penalizing the distances between features and their corresponding centers. Four-pointed stars indicate class centers. (c) Metric learning-based loss. Metric learning-based loss aims for both within-class compactness and between-class large-distance. (d) MLPNC loss. MLPNC loss combines class- and subclass-level positive–negative center losses, having coarse and fine discrimination ability. Four- and five-pointed stars indicate class centers and subclass centers, respectively.

In this article, ground truth similarities are still calculated by multiple experts (existing CNNs), which "score" similarity for each image pair. This is quite analogous to apparent age estimation, which also requires some experts (human beings) to "score" facial images in order to compute ground truth ages. Instead of working out a fixed apparent age for each facial image, Huo *et al.* [54] proposed estimating age distribution.

Inspired by Huo *et al.* [54], in this article, we consider the similarity between two images as a continuous random variable, whose ground truth probability density function and distribution function can be determined by experts, then CNNs are trained to learn similarity distribution. In this way, uncertainty as to similarity is fully taken into account, and the resulting similarity prediction is "continuous" instead of two-valued.

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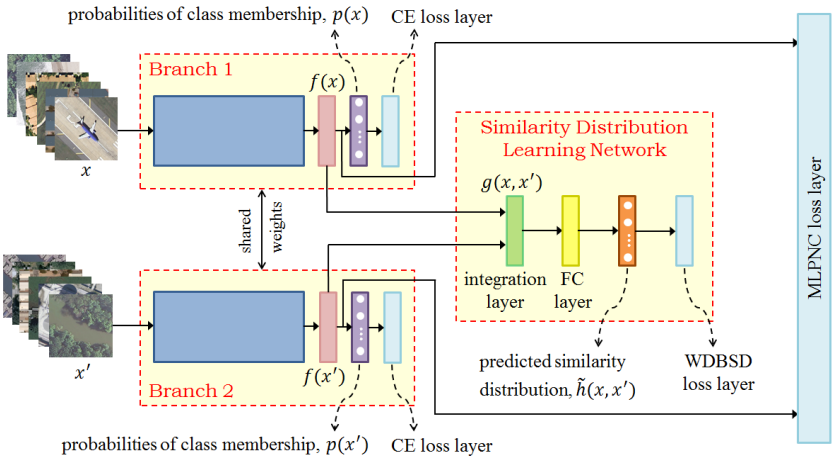


Fig. 4. Outline of network architecture. EM-CNN is composed of Siamese networks aimed at learning deep features and a similarity distribution learning network aimed at predicting similarity distribution for an image pair. EM-CNN is a three-task model, simultaneously conducting three tasks: center-metric learning, similarity distribution learning, and ASC.

In Sections IV-D and V, we show that similarity distribution learning is superior to both deep regression and ordinal classification in terms of performance.

3) Multitask Learning: Multitask learning improves generalization by leveraging the domain-specific information contained in the training signals of related tasks. It does this by training tasks in parallel while using a shared representation [55].

Multitask learning has been successfully applied to pedestrian detection [56], face recognition [57], person reidentification [58], scene classification [26], and so on, and it is usually found that learning multiple tasks jointly can lead to much performance improvement compared to learning them individually. Despite its success in various vision problems, multitask learning has not been fully studied in the remote sensing community.

The assumption underlying multitask learning is that all tasks, or at least, some of the tasks, are related to one another [59]. However, as pointed out in [55] and [59], one of the most important open problems in multitask learning is how to characterize, either formally or heuristically, what related tasks are, and the lack of an adequate definition of task relatedness is one of the obstacles preventing the development of more useful theories of multitask learning. Besides, it is claimed in [55], [59] that we may never have a theory of relatedness that allows us to reliably predict which tasks will help or hurt each other when trained simultaneously. Because of this, Caruana [55] suggested that people should focus part of their effort on ways of empirically determining which tasks are beneficially related to each other. Fortunately, existing studies have shown that the following tasks can benefit each other: 1) person verification and similarity learning [58] (intuitively, the former is somewhat similar to ASC), 2) query classification and web search [60] (intuitively, they are somewhat similar to ASC and CBAIR, respectively), and 3) ASC and contrastive embedding-based metric learning [26] (the latter, as well as our proposed center-metric learning, is a metric learning method). These promising results and our intuition encourage us to try multitask learning, although we cannot determine how much relatedness actually exists among our tasks.

Therefore, in this article, three tasks, that is, ASC, center-metric learning, and similarity distribution learning, are integrated into one CNN model. We find that our CNN, just like an eagle, can make fine visual discriminations, so we call it eagle-eyed multitask CNN (EM-CNN).

D. Contributions

Overall, our contributions are four-folded.

1) We propose a novel metric learning method called center-metric learning, which is developed from center learning and triplet embedding, not only inhering their advantages but also overcoming their weaknesses.
2) We construct a new kind of loss function called positive–negative center loss, including L1PNC loss capable of coarse discriminations, L2PNC loss capable of fine discriminations, and MLPNC loss capable of coarse and fine discriminations. To the best of our knowledge, this is the first time that deep fine-grained learning has been proposed in the remote sensing community.
3) We propose similarity distribution learning, elegantly embedding uncertainty regarding image similarity into the training process. As far as we know, this is the first time that similarity uncertainty has been aware of and dealt with.
4) We propose a model called EM-CNN, into which three tasks, that is, ASC, center-metric learning, and similarity distribution learning, are integrated. To our knowledge, this is the first time that CBAIR and ASC have been simultaneously handled.

II. Eagle-Eyed Multitask CNNs

This section delineates our proposed method.

The network architecture of EM-CNN is sketched out in Fig. 4. EM-CNN consists of two parts: the first part is Siamese networks aimed at learning deep features, whose two

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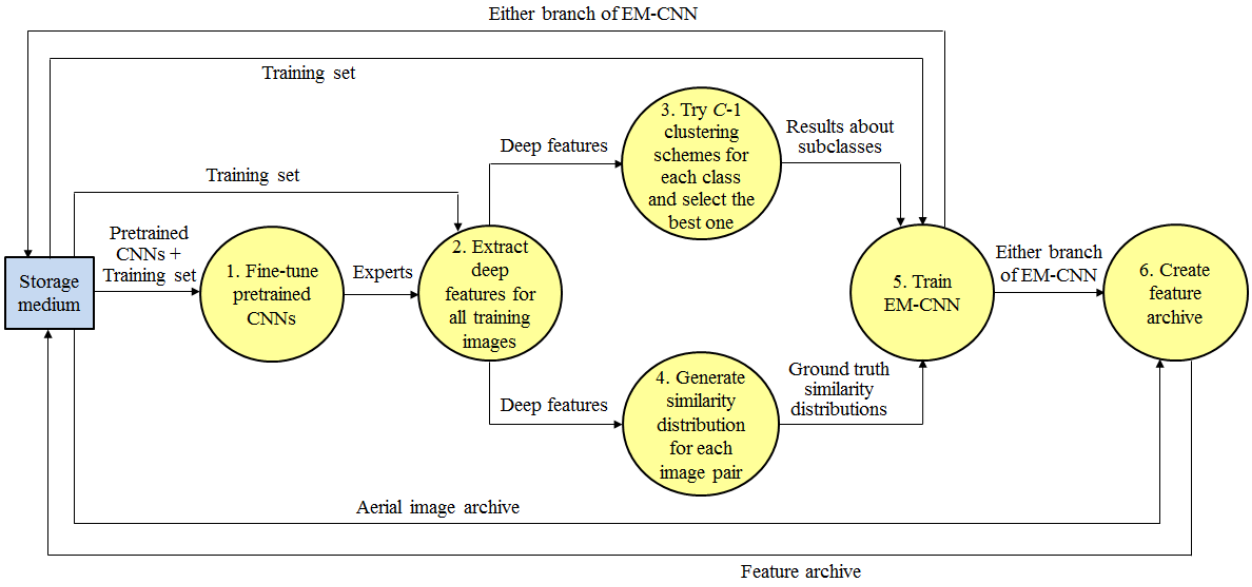


Fig. 5. Workflow. Main steps are numbered in sequence, 1, 2, 3, etc.

branches are identical in all aspects; the second part is a similarity distribution learning network aimed at predicting similarity distribution for an image pair.

As illustrated in Fig. 4, EM-CNN is coupled with three losses: CE loss aimed for ASC, MLPNC loss aimed for multiscale center-metric learning, and Wasserstein distance-based similarity distribution (WDBSD) loss aimed for similarity distribution learning. To "work" EM-CNN, we follow the workflow presented in Fig. 5. It primarily includes six stages:

1) M_1 ($M_1 \geq 4$) pretrained CNNs, the details of which can be found in Section III-B, are fine-tuned using aerial training images. We refer to the resulting models as experts or CNN experts.
2) Deep features for all training images are extracted from experts (these features, aimed at the subsequent subclass generation and similarity distribution creation, are "intermediate products" and not "final products").
3) Deep features are used to cluster training images from each class into c subclasses ($c = 2, 3, \ldots, C$; $C \geq 2$ is a given constant), and then the optimal value of c is determined according to a preestablished criterion. This step makes preparations for computing L2PNC loss, which is an important constituent of MLPNC loss and should be computed based on subclass membership.
4) Deep features are used to create ground truth similarity distribution for each image pair. This step makes preparations for computing WDBSD loss, which measures the difference between ground truth and predicted similarity distributions.
5) EM-CNN is trained: Labeled training images, clustering results generated at stage 3, and similarity distributions yielded at stage 4 are employed to compute the gradients of CE loss, MLPNC loss, and WDBSD loss, providing a basis for optimization.
6) From the trained EM-CNNs we detach either branch and utilize it to extract final deep features for all images in the aerial image archive. Then these FVs are stored in the feature archive for later use.

In what follows, we elaborate on our methodology.

A. Creating Subclasses

At the first stage, fine-tuning yields M_1 experts, all of "whom" participate in working out similarity distributions (see Section II-C) to guarantee statistically reliable estimates. Our subclasses/clusters are created through clustering. Since clustering does not involve statistical inference, fewer experts will suffice. Therefore, from among M_1 experts, M_2 ones are selected ($M_2 \leq M_1$). They serve as feature extractors and provide the basis for clustering.

Given a training image x, its M_2 FVs are L_2-normalized and then concatenated. We can consider that there exists a function $\ddot{f}$ that maps every x to a concatenated FV. So, we denote by $\ddot{f}(x)$ the concatenated vector, whose subclass membership, after clustering finishes, determines which subclass x belongs to.

K-means [61] acts as our clustering algorithm. We denote by $\mathscr{D}$ an image set composed of the training images that belong to a given class, and by c the number of subclasses. With c varying from 2 to C, $(C-1)$ clustering schemes are tried, the $(c-1)$th one partitioning $\mathscr{D}$ into c clusters.

We utilize the Calinski–Harabasz criterion (CHC), which is best suited for K-means clustering solutions with squared Euclidean distances [62], to select the best clustering scheme. Let

$$\mathscr{D} = \bigcup_{i=1}^{c} \mathscr{D}_c^i, \quad c = 2, 3, \ldots, C \tag{6}$$

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where $\mathscr{D}_c^i$ is the ith image subset (i.e., subclass i) in the c-subclass scheme, and let

$$\Xi = \frac{1}{|\mathscr{D}|}\sum_{x\in\mathscr{D}} \ddot{f}(x) \tag{7}$$

and

$$\xi_c^i = \frac{1}{\left|\mathscr{D}_c^i\right|}\sum_{x\in\mathscr{D}_c^i} \ddot{f}(x), \quad c = 2, 3, \ldots, C;\ \ i = 1, 2, \ldots, c \tag{8}$$

where $|\cdot|$ indicates the size of an image set. Then the overall between-cluster variance is defined as

$$S_c^{\mathrm{B}} = \sum_{i=1}^{c}\left|\mathscr{D}_c^i\right|\left\|\xi_c^i - \Xi\right\|_2^2, \quad c = 2, 3, \ldots, C \tag{9}$$

and the overall within-cluster variance is defined as

$$S_c^{\mathrm{W}} = \sum_{i=1}^{c}\sum_{x\in\mathscr{D}_c^i}\left\|\ddot{f}(x) - \xi_c^i\right\|_2^2, \quad c = 2, 3, \ldots, C. \tag{10}$$

Obviously, well-defined clusters have a large S_c^{B} and a small S_c^{W}. Then, the Calinski–Harabasz index (CHI) is defined as

$$\mathrm{CHI}_c = \frac{(|\mathscr{D}| - c)S_c^{\mathrm{B}}}{(c-1)S_c^{\mathrm{W}}}, \quad c = 2, 3, \ldots, C. \tag{11}$$

The larger the CHI, the better the data split. Maximizing CHI_c with respect to c, we have

$$n = \arg\max_{c\in\{2,3,\ldots,C\}} \mathrm{CHI}_c \tag{12}$$

this is the optimal number of subclasses for the given class.

Besides CHC, another two clustering evaluation criteria, Davies–Bouldin Criterion (DBC) [63] and silhouette criterion (SC) [64], are also used to determine the optimal number of clusters, n. We make an experimental comparison of these criteria in Section IV-B.

Hereafter, multiple classes are involved, so we add a subscript to differentiate their numbers of subclasses. That is to say, class j has n_j subclasses ($j = 1, 2, \ldots, N$).

B. Center-Metric Learning

Center loss (5) enables CNNs to learn N class centers. For a training image x, we call $\Theta_{Y(x)}$ the positive center of x, and Θ_j $[j = 1, 2, \ldots, N;\ \ j \neq Y(x)]$ the negative centers of x. Let

$$Z(x) = \arg\min_{\substack{j\in\{1,2,\ldots,N\}\\ j\neq Y(x)}} \left\|f(x) - \Theta_j\right\|_2^2 \tag{13}$$

obviously, $\Theta_{Z(x)}$ is the nearest negative center of x, and let

$$D(x) = \left\|f(x) - \Theta_{Y(x)}\right\|_2^2 - \left\|f(x) - \Theta_{Z(x)}\right\|_2^2. \tag{14}$$

For a given mini-batch $\mathscr{B}$, we define L1PNC loss as

$$\mathcal{L}_{\mathrm{l1pnc}}(\omega; \mathscr{B}) = \frac{1}{2}\sum_{x\in\mathscr{B}} \max(D(x) + T, 0) \tag{15}$$

where $T \geq 0$ is a margin threshold. L1PNC loss tries to simultaneously pull $f(x)$ toward the positive center and push $f(x)$ away from all negative centers, aiming for a marginal distance difference, T. That is, penalty is imposed until the "positive distance" "beats" the smallest "negative distance" by a margin of T. In contrast to center loss, which only focuses on within-class closeness, L1PNC loss copes with both within-class pulling and across-class pushing. Coupled with triplets, each of which consists of $f(x)$, $\Theta_{Y(x)}$, and $\Theta_{Z(x)}$, L1PNC loss bears a striking resemblance to triplet loss [2], [51].

Denote by θ_j^i the ith subclass center of class j, and by $y(x)$ the subclass label of x. Then, the subclass-level counterparts of (13) and (14) can be given as follows:

$$z(x) = \arg\min_{\substack{i\in\{1,2,\ldots,n_{Y(x)}\}\\ i\neq y(x)}} \left\|f(x) - \theta_{Y(x)}^i\right\|_2^2 \tag{16}$$

and

$$d(x) = \left\|f(x) - \theta_{Y(x)}^{y(x)}\right\|_2^2 - \left\|f(x) - \theta_{Y(x)}^{z(x)}\right\|_2^2. \tag{17}$$

Similarly, L2PNC loss is defined as

$$\mathcal{L}_{\mathrm{l2pnc}}(\omega; \mathscr{B}) = \frac{1}{2}\sum_{x\in\mathscr{B}} \max(d(x) + t, 0) \tag{18}$$

where $t \geq 0$ is a margin threshold. L2PNC loss is a fine-grained version of L1PNC loss, aiming for local pulling–pushing and compelling CNNs to make fine visual discriminations.

Combining L1PNC loss with L2PNC loss, we obtain the MLPNC loss

$$\mathcal{L}_{\mathrm{mlpnc}}(\omega; \mathscr{B}) = \mathcal{L}_{\mathrm{l1pnc}}(\omega; \mathscr{B}) + \mathcal{L}_{\mathrm{l2pnc}}(\omega; \mathscr{B}). \tag{19}$$

Let

$$\mathcal{C}(x) = \frac{1}{2}[\max(D(x) + T, 0) + \max(d(x) + t, 0)]. \tag{20}$$

Using (15) and (18)-(20), we can rewrite MLPNC loss as

$$\mathcal{L}_{\mathrm{mlpnc}}(\omega; \mathscr{B}) = \sum_{x\in\mathscr{B}} \mathcal{C}(x). \tag{21}$$

MLPNC loss is capable of fine and coarse discrimination of images and, hence, endowing our CNN with keen eyesight.

L1PNC, L2PNC, and MLPNC losses are collectively referred to as positive–negative center loss; and a learning process equipped with positive–negative center loss is referred to as center-metric learning, which blends center learning with metric learning.

Center-metric learning does not need any predesigned image pairs/triplets; instead, all triplets, each of which comprises a FV, a positive center, and a nearest negative center, are automatically learned in the training process. Furthermore, our number of (sub)class-level triplets is exactly equal to that of input images, and hard sample mining is neatly sidestepped. So center-metric learning stands in marked contrast to other CNN-based metric learning methods [2], [26], [50], [51], which depend heavily on a tremendous number of carefully designed image pairs/triplets and face the thorny problem of treating easy and hard samples differently.

C. Similarity Distribution Learning

M_1 CNN experts, a detailed introduction to "whom" can be found in Section III-B, are responsible for computation of similarity distributions. Based on the activations extracted

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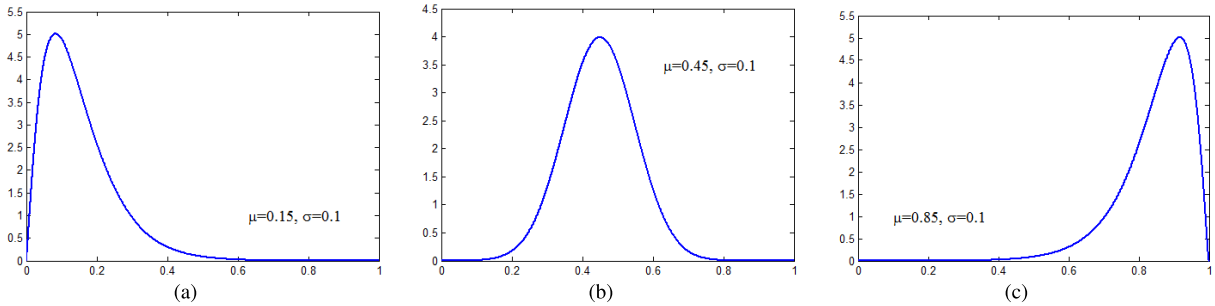


Fig. 6. Probability density functions. (a) Gamma probability density function ($\mu = 0.15$, $\sigma = 0.1$). (b) Gaussian probability density function ($\mu = 0.45$, $\sigma = 0.1$). (c) Vertically flipped Gamma probability density function ($\mu = 0.85$, $\sigma = 0.1$).

from experts' penultimate layers (i.e., the layers before the class membership prediction layers), Euclidean distance and similarity between two images can be calculated, so we say experts "compute" Euclidean distance and "score" similarity for each image pair.

Given a training set $\mathscr{T}$, expert m ($m = 1, 2, \ldots, M_1$) computes Euclidean distances for all image pairs and normalizes them into $[0, 1]$. Denote by $\langle x, x' \rangle$ and e_m an image pair and its normalized Euclidean distance computed by expert m, respectively, then its similarity scored by expert m is

$$s_m = 1 - e_m, \quad m = 1, 2, \ldots, M_1 \tag{22}$$

obviously, $s_m \in [0, 1]$.

We consider the similarity of $\langle x, x' \rangle$ to be a continuous random variable $\boldsymbol{S}$, whose expected value and standard deviation are, respectively, given by

$$\mu = \frac{1}{M_1} \sum_{m=1}^{M_1} s_m \tag{23}$$

and

$$\sigma = \sqrt{\frac{1}{M_1} \sum_{m=1}^{M_1} (s_m - \mu)^2}. \tag{24}$$

We hope to find an appropriate probability distribution model for $\boldsymbol{S}$. In image esthetics analysis, Murray *et al.* [65] found that for the images with mean human ratings between 2 and 8 (note that human esthetics ratings range from 0 to 10), esthetics scores can be modeled by Gaussian distributions, whereas for those with highly skewed mean human ratings, esthetics scores can be statistically approximated by Gamma distributions. Similar findings have been reported in image quality assessment [66]. Inspired by these results, we consider that $\boldsymbol{S}$ obeys the normal distribution when $\mu \in [0.2, 0.8]$, and $\boldsymbol{S}$ obeys a Gamma distribution when $\mu \in [0, 0.2) \cup (0.8, 1.0]$.

The shape parameter and scale parameter of Gamma distribution are, respectively, defined as [61]

$$\alpha(\mu, \sigma) = \frac{\mu^2}{\sigma^2} \tag{25}$$

and

$$\beta(\mu, \sigma) = \frac{\sigma^2}{\mu}. \tag{26}$$

Let

$$\psi(s; \mu, \sigma) = \frac{s^{\alpha(\mu,\sigma)-1} \exp\left(-\frac{s}{\beta(\mu,\sigma)}\right)}{[\beta(\mu, \sigma)]^{\alpha(\mu,\sigma)} \Gamma(\alpha(\mu, \sigma))} \tag{27}$$

and denote the probability density function of $\boldsymbol{S}$ as $q(s; \mu, \sigma)$, which is defined as follows.

1) When $\mu \in [0, 0.2)$,

$$q(s; \mu, \sigma) = \psi(s; \mu, \sigma). \tag{28}$$

2) When $\mu \in [0.2, 0.8]$,

$$q(s; \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(s-\mu)^2}{2\sigma^2}\right). \tag{29}$$

3) When $\mu \in (0.8, 1.0]$,

$$q(s; \mu, \sigma) = \psi(1 - s; 1 - \mu, \sigma). \tag{30}$$

Fig. 6 displays three probability density functions corresponding to the three cases described above, respectively.

Since CNNs cannot deal with continuous data, we need to produce a discretized form of similarity distribution. To this end, we split the interval $[0, 1]$ into L equal subintervals ($L \geq 2$), and let

$$\begin{aligned} h_l = &\int_{(l-1)/L}^{l/L} q(s; \mu, \sigma) \mathrm{d}s \\ &+ \frac{1}{L} \left[\int_{-\infty}^{0} q(s; \mu, \sigma) \mathrm{d}s + \int_{1}^{+\infty} q(s; \mu, \sigma) \mathrm{d}s \right] \\ &l = 1, 2, \ldots, L. \end{aligned} \tag{31}$$

Obviously, $h_l \geq 0$ ($l = 1, 2, \ldots, L$). Note that the second term on the right-hand side is added to make sure $\sum_{l=1}^{L} h_l = 1$. We refer to

$$h = (h_1, h_2, \ldots, h_L)^{\mathbf{T}} \tag{32}$$

as the ground truth similarity distribution of $\langle x, x' \rangle$. In fact, h can be considered as a bivariate vector function of x and x'. For consistency of notation, hereafter we use $h(x, x')$ instead.

Similarity distribution learning network, which is made up of an integration layer, an FC layer, an output layer, and a loss layer, aims to predict similarity distributions. As illustrated in Fig. 4, $f(x)$ (the learned features of x) and $f(x')$ (the learned features of x') are consolidated in the integration layer, yielding $g(x, x')$:

$$g(x, x') = [f(x) - f(x')] \cdot *[f(x) - f(x')] \tag{33}$$

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where "$\cdot*$" means element-by-element multiplication. The integration layer ensures that swapping x and x' does not affect predictions. After passing through the FC layer and then being softmax-processed, $g(x, x')$ is transformed into an L-dimensional vector, $\tilde{h}(x, x')$, which satisfies $\tilde{h}_l(x, x') \geq 0$ $(l = 1, 2, \ldots, L)$ and $\sum_{l=1}^{L} \tilde{h}_l(x, x') = 1$ due to the softmax operations. We call $\tilde{h}(x, x')$ the predicted similarity distribution of $\langle x, x' \rangle$.

Then, Wasserstein distance [1], [29], [67], which is among the most well-known methods for measuring the difference between two probability distributions, is used to compute the distance between $h(x, x')$ and $\tilde{h}(x, x')$

$$\mathcal{W}(x, x') = \frac{1}{2} \sum_{\hat{l}=1}^{L-1} \left\{ \sum_{l=1}^{\hat{l}} \left[h_l(x, x') - \tilde{h}_l(x, x') \right] \right\}^2. \quad (34)$$

In the optimization process, training images are paired (see Fig. 4), forming many mini-batches of image pairs. We denote any mini-batch of image pairs as $\langle \mathscr{B}, \mathscr{B}' \rangle$, with $\mathscr{B}$ fed to branch 1 and $\mathscr{B}'$ fed to branch 2. Note that $\left| \langle \mathscr{B}, \mathscr{B}' \rangle \right| = |\mathscr{B}| = \left| \mathscr{B}' \right|$, where $\left| \langle \mathscr{B}, \mathscr{B}' \rangle \right|$ means the number of image pairs in $\langle \mathscr{B}, \mathscr{B}' \rangle$. Then we define WDBSD loss as

$$\mathcal{L}_{\text{wdbsd}}(\omega; \mathscr{B}, \mathscr{B}') = \sum_{\langle x, x' \rangle \in \langle \mathscr{B}, \mathscr{B}' \rangle} \mathcal{W}(x, x'). \quad (35)$$

WDBSD loss gives full consideration to uncertainty about similarity by forcing EM-CNN to learn similarity distributions, and implants multiple-CNN information into EM-CNN by minimizing the divergence between predicted similarity distributions and ground truth similarity distributions, which are provided by M_1 CNN experts.

Furthermore, the predicted SS of $\langle x, x' \rangle$ is given by

$$\bar{s}(x, x') = \sum_{l=1}^{L} \frac{2l - 1}{2L} \tilde{h}_l(x, x') \quad (36)$$

where $((2l-1)/2L)$ is the median value of the lth subinterval $[(l - 1/L), l/L]$.

Apparently, $\bar{s}(x, x')$ can take any value in [0, 1], unambiguously telling us how similar two images are, thereby paving the way for fine-grained CBAIR. This is in stark contrast to existing similarity learning models [46], [47], which, with a two-valued prediction scheme, are only capable of determining whether or not two images are similar.

After training is finished, either branch is detached from EM-CNN and used to create a feature archive, which should be saved for later online retrieval. When users provide a query, it is fed to the branch to generate a FV, which is then paired with all vectors in the feature archive. Subsequently, FV pairs are input to the similarity distribution learning network of EM-CNN to generate the predicted similarity distributions. Then, the predicted SSs are computed using (36). Finally, SSs are sorted in descending order, and the corresponding high-ranked aerial images are presented to the users. This procedure is illustrated in Fig. 7.

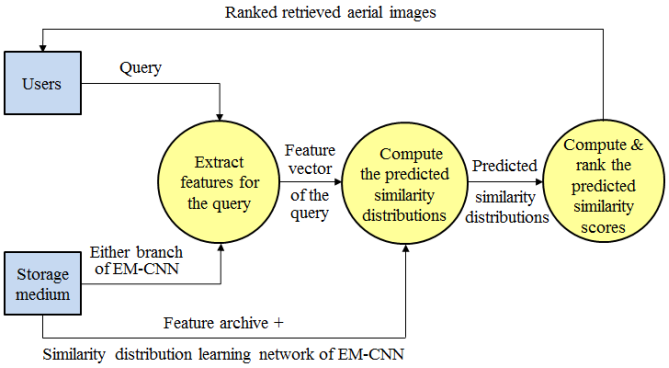


Fig. 7. Online retrieval process (figure adapted from [1]). A query is fed to either branch of EM-CNN to yield a FV, which is then paired with all vectors in the feature archive. Then, FV pairs are input to the similarity distribution learning network to generate the predicted similarity distributions. Subsequently, the predicted SSs are calculated and sorted in descending order. Finally, the corresponding high-ranked aerial images are presented to the users.

D. Eagle-Eyed Multitask CNNs

As shown in Fig. 4, each branch of EM-CNN has a CE loss layer, which is aimed at penalizing misclassification. We incorporate CE loss, MLPNC loss, and WDBSD loss into the CE-MLPNC-WDBSD (C-M-W) loss

$$\mathcal{L}_{\text{c-m-w}}(\omega; \mathscr{B}, \mathscr{B}') = \mathcal{L}_{\text{ce}}(\omega; \mathscr{B} \cup \mathscr{B}') + \lambda_1 \mathcal{L}_{\text{mlpnc}}(\omega; \mathscr{B} \cup \mathscr{B}') + \lambda_2 \mathcal{L}_{\text{wdbsd}}(\omega; \mathscr{B}, \mathscr{B}') \quad (37)$$

where $\lambda_1 \geq 0$ and $\lambda_2 \geq 0$ are tunable parameters, which determine the tradeoff among three kinds of losses.

Equipped with C-M–W loss, EM-CNN is trained to simultaneously carry out three tasks, that is, ASC, center-metric learning, and similarity distribution learning. Hopefully, these three tasks will complement one another, thereby endowing EM-CNN with a better generalization ability. In particular, MLPNC loss enables EM-CNN to recognize within-class variations, and WDBSD loss further strengthens the fine discrimination ability of EM-CNN.

Moreover, it should be stressed that although training images must be input to EM-CNN in pairs, image pairing is aimed only at similarity distribution learning, and we develop a simple, yet effective similarity-specific method for generating image pairs (see Section III-C). As for center-metric learning, it has nothing to do with the Siamese structure of EM-CNN, and can be applied to any network architecture, including single-branch CNNs.

During training, network parameters ω, class centers $\{\Theta_j | j = 1, 2, \ldots, N\}$, and subclass centers $\{\theta_j^i | j = 1, 2, \ldots, N;\ i = 1, 2, \ldots, n_j\}$ should be iteratively updated. As indicated in Fig. 4, ω consists of three parts: the similarity distribution learning network's parameters, the branches' output layers' parameters (the output layers are colored purple), and the remaining branch parameters; we denote them as ω^S (S means "similarity"), ω^{OB} (OB means "the output layers of branches"), and ω^{RB} (RB means "the remaining layers of branches"), respectively.

SGD [48] serves as our optimization algorithm. Since $\mathcal{L}_{\text{ce}}(\omega; \mathscr{B} \cup \mathscr{B}')$ and $\mathcal{L}_{\text{mlpnc}}(\omega; \mathscr{B} \cup \mathscr{B}')$ are not (composite) functions of ω^S, the gradients of $\mathcal{L}_{\text{c-m-w}}(\omega; \mathscr{B}, \mathscr{B}')$ with

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respect to ω^S can be computed as follows:

$$\frac{\partial \mathcal{L}_{\text{c-m-w}}\left(\omega; \mathscr{B}, \mathscr{B}'\right)}{\partial \omega^S} = \lambda_2 \frac{\partial \mathcal{L}_{\text{wdbsd}}\left(\omega; \mathscr{B}, \mathscr{B}'\right)}{\partial \omega^S}. \tag{38}$$

Note that ω^S is composed of parameters of multiple layers and, hence, (38) is only a schematic formula. We may need to apply the chain rule when computing gradients for individual layers.

Analogously, the gradients with respect to ω^{OB} are

$$\frac{\partial \mathcal{L}_{\text{c-m-w}}\left(\omega; \mathscr{B}, \mathscr{B}'\right)}{\partial \omega^{\text{OB}}} = \frac{\partial \mathcal{L}_{\text{ce}}\left(\omega; \mathscr{B} \cup \mathscr{B}'\right)}{\partial \omega^{\text{OB}}}. \tag{39}$$

From (4), (21), (35), and (37), it follows that

$$\begin{aligned}
&\frac{\partial \mathcal{L}_{\text{c-m-w}}\left(\omega; \mathscr{B}, \mathscr{B}'\right)}{\partial \omega^{\text{RB}}} \\
&= \frac{\partial \mathcal{L}_{\text{ce}}\left(\omega; \mathscr{B} \cup \mathscr{B}'\right)}{\partial \omega^{\text{RB}}} + \lambda_1 \frac{\partial \mathcal{L}_{\text{mlpnc}}\left(\omega; \mathscr{B} \cup \mathscr{B}'\right)}{\partial \omega^{\text{RB}}} \\
&\quad + \lambda_2 \frac{\partial \mathcal{L}_{\text{wdbsd}}\left(\omega; \mathscr{B}, \mathscr{B}'\right)}{\partial \omega^{\text{RB}}} \\
&= \sum_{x \in \mathscr{B} \cup \mathscr{B}'} \frac{\partial \mathcal{E}(x)}{\partial f(x)} \frac{\partial f(x)}{\partial \omega^{\text{RB}}} + \lambda_1 \sum_{x \in \mathscr{B} \cup \mathscr{B}'} \frac{\partial \mathcal{C}(x)}{\partial f(x)} \frac{\partial f(x)}{\partial \omega^{\text{RB}}} \\
&\quad + \lambda_2 \sum_{\substack{\langle x, x' \rangle \in \\ \langle \mathscr{B}, \mathscr{B}' \rangle}} \frac{1}{2} \left[\frac{\partial \mathcal{W}(x, x')}{\partial f(x)} \frac{\partial f(x)}{\partial \omega^{\text{RB}}} + \frac{\partial \mathcal{W}(x, x')}{\partial f(x')} \frac{\partial f(x')}{\partial \omega^{\text{RB}}} \right].
\end{aligned} \tag{40}$$

Note that $f(x)$ can be seen as a function of ω ("ω" is dropped to simplify the notation), so $(\partial f(x)/\partial \omega^{\text{RB}})$ and $(\partial f(x')/\partial \omega^{\text{RB}})$ in (40) are nontrivial.

Then, the parameters of EM-CNN can be updated in the following order:

$$\omega^S \leftarrow \omega^S - \rho \frac{\partial \mathcal{L}_{\text{c-m-w}}\left(\omega; \mathscr{B}, \mathscr{B}'\right)}{\partial \omega^S} \tag{41}$$

$$\omega^{\text{OB}} \leftarrow \omega^{\text{OB}} - \rho \frac{\partial \mathcal{L}_{\text{c-m-w}}\left(\omega; \mathscr{B}, \mathscr{B}'\right)}{\partial \omega^{\text{OB}}} \tag{42}$$

and

$$\omega^{\text{RB}} \leftarrow \omega^{\text{RB}} - \rho \frac{\partial \mathcal{L}_{\text{c-m-w}}\left(\omega; \mathscr{B}, \mathscr{B}'\right)}{\partial \omega^{\text{RB}}} \tag{43}$$

where $\rho > 0$ is the learning rate.

After network parameter updating has been finished, class centers and subclass centers are updated, preparing for the next iteration.

Let

$$A_j(x) = D(x) + T > 0 \,\&\, Y(x) = j, \quad j = 1, 2, \ldots, N \tag{44}$$

and

$$B_j(x) = D(x) + T > 0 \,\&\, Z(x) = j, \quad j = 1, 2, \ldots, N \tag{45}$$

note that the right-hand sides of (44) and (45) are logic expressions, each of which evaluates to either "true" or "false."

Using (14) and (15), we can compute the average gradient of $\mathcal{L}_{\text{l1pnc}}\left(\omega; \mathscr{B} \cup \mathscr{B}'\right)$ with respect to Θ_j as follows (note that CE loss, WDBSD loss and L2PNC loss have nothing to do with class centers)

$$\begin{aligned}
\Delta_j = &\frac{\sum_{x \in \mathscr{B} \cup \mathscr{B}'} \left[\Theta_{Y(x)} - f(x)\right] \zeta(A_j(x))}{\epsilon + \sum_{x \in \mathscr{B} \cup \mathscr{B}'} \zeta(A_j(x))} \\
&+ \frac{\sum_{x \in \mathscr{B} \cup \mathscr{B}'} \left[f(x) - \Theta_{Z(x)}\right] \zeta(B_j(x))}{\epsilon + \sum_{x \in \mathscr{B} \cup \mathscr{B}'} \zeta(B_j(x))}, \quad j = 1, 2, \ldots, N
\end{aligned} \tag{46}$$

where ϵ is a very small positive number guaranteeing that the denominators are not equal to 0 in any case, and

$$\zeta(\text{logic expression}) = \begin{cases} 1, & \text{if logic expression is true} \\ 0, & \text{if logic expression is false.} \end{cases} \tag{47}$$

Then the updating rule of class centers is

$$\Theta_j \leftarrow \Theta_j - R\Delta_j, \quad j = 1, 2, \ldots, N \tag{48}$$

where $R > 0$ is the center learning rate aiming to avoid large perturbations caused by the few possibly mislabeled samples [49].

Let

$$\begin{aligned}
a_j^i(x) = d(x) + t > 0 \,\&\, Y(x) = j \,\&\, y(x) = i \\
j = 1, 2, \ldots, N; \quad i = 1, 2, \ldots, n_j
\end{aligned} \tag{49}$$

and

$$\begin{aligned}
b_j^i(x) = d(x) + t > 0 \,\&\, Y(x) = j \,\&\, z(x) = i \\
j = 1, 2, \ldots, N; \; i = 1, 2, \ldots, n_j
\end{aligned} \tag{50}$$

then, in the same way as described for the coarse level, subclass centers can be updated using

$$\theta_j^i \leftarrow \theta_j^i - r\delta_j^i, \quad j = 1, 2, \ldots, N; \; i = 1, 2, \ldots, n_j \tag{51}$$

where $r > 0$ and

$$\begin{aligned}
\delta_j^i = &\frac{\sum_{x \in \mathscr{B} \cup \mathscr{B}'} \left[\theta_{Y(x)}^{y(x)} - f(x)\right] \zeta(a_j^i(x))}{\epsilon + \sum_{x \in \mathscr{B} \cup \mathscr{B}'} \zeta(a_j^i(x))} \\
&+ \frac{\sum_{x \in \mathscr{B} \cup \mathscr{B}'} \left[f(x) - \theta_{Y(x)}^{z(x)}\right] \zeta(b_j^i(x))}{\epsilon + \sum_{x \in \mathscr{B} \cup \mathscr{B}'} \zeta(b_j^i(x))}
\end{aligned} \tag{52}$$

is the average gradient of $\mathcal{L}_{\text{l2pnc}}\left(\omega; \mathscr{B} \cup \mathscr{B}'\right)$ with respect to θ_j^i.

III. Experimental Setup and Parameter Selection

This section presents the experimental setup first, and then details how hyper-parameters are determined.

A. Data Sets

Our experiments involve four challenging aerial image sets: PatternNet [36], UC-Merced (UCM) [68], NWPU-RESISC45 (NR) [11], and Aerial Image Data set (AID) [11]. The first two data sets are used for CBAIR, and the latter two are used for ASC.

PatternNet has 38 classes, each including 800 images, and the spatial resolution is as high as 0.062–4.693 m. UCM contains 21 categories and 2100 images, with 100 images

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Fig. 8. Example images from UCM. (1) Agricultural. (2) Airplane. (3) Baseball diamond. (4) Beach. (5) Building. (6) Chaparral. (7) Dense residential. (8) Forest. (9) Freeway. (10) Golf course. (11) Harbor. (12) Intersection. (13) Medium residential. (14) Mobile home park. (15) Overpass. (16) Parking lot. (17) River. (18) Runway. (19) Sparse residential. (20) Storage tank. (21) Tennis court.

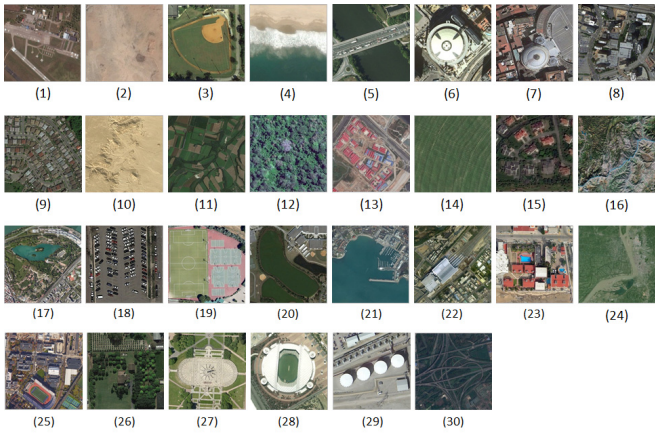


Fig. 9. Example images from AID. (1) Airport. (2) Bare land. (3) Baseball field. (4) Beach. (5) Bridge. (6) Center. (7) Church. (8) Commercial. (9) Dense residential. (10) Desert. (11) Farmland. (12) Forest. (13) Industrial. (14) Meadow. (15) Medium residential. (16) Mountain. (17) Park. (18) Parking. (19) Playground. (20) Pond. (21) Port. (22) Railway station. (23) Resort. (24) River. (25) School. (26) Sparse residential. (27) Square. (28) Stadium. (29) Storage tank. (30) Viaduct.

per class, and the spatial resolution is 0.3 m per pixel. NR consists of 31 500 images coming from 45 scene categories, and the spatial resolution ranges from 30 to 0.2 m. As for AID, it has 10 000 images, which cover 30 categories, and the spatial resolution varies from 8 to 0.5 m. Except for AID, whose image size is 600 × 600, every data set comprises images of size 256 × 256. Some examples from UCM and AID are presented in Figs. 8 and 9, respectively.

PatternNet, UCM, and AID are randomly split into subsets for training and test, with training/test ratios being 20%/80%, 50%/50%, and 50%/50%, respectively. The largest NR provides a validation set for us to select the hyper-parameters (see Section III-D), and its training/validation/test ratio is 20%/10%/70%.

Furthermore, we conduct data augmentation. The expanded training sets of PatternNet, NR, and AID are 12 times larger than the original sizes: five subimages with 75% coverage are cut out from corners and center, and then both the subimages and the original image are flipped vertically or horizontally with a probability of 0.5. Since UCM is too small, its training set is expanded to 20 times the original size: four additional subimages with 50% coverage are cut out from corners and then randomly flipped.

TABLE I
DETAILS OF DATA SETS

Dataset	Purpose	# Class	# Samples per Class	# Total Samples	# Training Samples	# Validation Samples	# Test Samples
PatternNet	CBAIR	38	800	30.4 K	72.96 K (A)[1]	-	24.32 K
UCM	CBAIR	21	100	2.1 K	21 K (A)[1]	-	1.05 K
NR	ASC	45	700	31.5 K	75.6 K (A)[1]	3.15 K	22.05 K
AID	ASC	30	Unequal	10.0 K	60.0 K (A)[1]		5.0 K

[1] *A* means "augmented."

It should be pointed out that some evaluation measures, for example, P@k (see Section IV-A), depend on test set size. Over PatternNet, most of the previous studies [1], [34], [36], [40], [44] used 20% data for the test; therefore, for the sake of fair comparison, we randomly divide our PatternNet test set into four equal parts, compute evaluation measures over each part separately, and report the average values.

The details of the data sets are shown in Table I.

B. CNN Experts

We set $M_1 = 12$ in our experiments, that is, 12 CNN experts are used.

We select 12 well-known pretrained CNNs: CaffeRef [69], AlexNet [70], VGG-S [71], VGG-F [71], VGG-M [71], VGG-VD-16 [13], VGG-VD-19 [13], ResNet-50 [72], ResNet-101 [72], ResNet-152 [72], NetVLAD [73], and PlacesNet [74]. Except NetVLAD that was pretrained for place recognition using panoramic street-view images and PlacesNet that was pretrained for scene classification using natural scene images, all models were pretrained over the famous ImageNet [8], an everyday image set.

As described in Section II, 12 pretrained CNNs are fine-tuned over aerial image sets, then the resulting experts serve as feature generators and provide the basis for computation of similarity distributions. The details of feature extraction are summarized in Table II.

As for the value of M_2, as stated in Section II-A, $M_2 < 12$ can satisfy the requirements for clustering, because clustering process has no connection with statistical inference. Using more experts means a heavier computation burden, although it leads to a better performance frequently [1], [29]. Besides, in [29], multiple experts are used to aggregate images from different semantic classes and create a binary category tree, and it is found that three experts will suffice to achieve satisfactory results. Taking into consideration these findings as well as the tradeoff between performance improvement and time/space consumption, we set $M_2 = 3$.

In addition, the 12 experts can be categorized into 3 CNN families: 1) the CaffeRef-type family, including experts 1, 4, 8, 2) the VGG family, including experts 2, 5, 7, 9, 11, 12, and 3) the ResNet family, including experts 3, 6, 10. Because CNNs from different families complement one another better

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TABLE II
DETAILS OF EXPERTS

No.	CNN Experts	Layer Used	Feature Dimension
1	CaffeRef [69]	The 2nd FC layer	4096
2	VGG-VD-16 [13]	The 2nd FC layer	4096
3	ResNet-50 [72]	The 5th pooling layer	2048
4	AlexNet [70]	The 2nd FC layer	4096
5	VGG-S [71]	The 2nd FC layer	4096
6	ResNet-101 [72]	The 5th pooling layer	2048
7	VGG-F [71]	The 2nd FC layer	4096
8	PlacesNet [74]	The 2nd FC layer	4096
9	VGG-VD-19 [13]	The 2nd FC layer	4096
10	ResNet-152 [72]	The 5th pooling layer	2048
11	VGG-M [71]	The 2nd FC layer	4096
12	NetVLAD [73]	The NetVLAD layer	4096

than those from the same family [1], [10], [11], [29], we select one expert from each family. More specifically, the first three experts in Table II, that is, CaffeRef, VGG-VD-16, and ResNet-50, are utilized to create subclasses.

C. Generation of Image Pairs

During training, images input to EM-CNN must be in pairs for the purpose of computation of similarity distributions. As in [1], we develop a simple, yet effective algorithm to generate image pairs.

Since multiple image pairs are involved, we change the notation "μ" in (23) to "$\mu(x, x')$" to clarify that it is coupled with the image pair $\langle x, x' \rangle$. To simplify notation, we let x and x' signify image numbers in this section, hopefully without causing confusion. Given an enlarged training set $\mathscr{T}$, we first create a $|\mathscr{T}| \times |\mathscr{T}|$ matrix U whose (x, x') entry is $\mu(x, x')$ computed by (23), then a set of image pairs, $\mathscr{S}$, can be formed as follows:

$\mathscr{S} = \emptyset$
for $x = 1 : |\mathscr{T}|$
  Divide the nonnegative entries in U's xth row into 20 parts such that the ith part comprises the entries falling within $[0.05(i-1), 0.05i]$
  for $i = 1 : 20$
    Randomly select one entry $\mu(x, x')$ from the ith part
    Add the image pair $\langle x, x' \rangle$ to $\mathscr{S}$
    $\mu(x', x) = -1$
  end
end

This algorithm is easy to implement, steering clear of the troubles usually going with image pair/triplet generation. Furthermore, the resulting image pairs have SSs that are basically evenly distributed in the interval $[0, 1]$, ensuring the completeness and diversity of training samples. Besides, the algorithm guarantees that image pairs are not repeatedly selected.

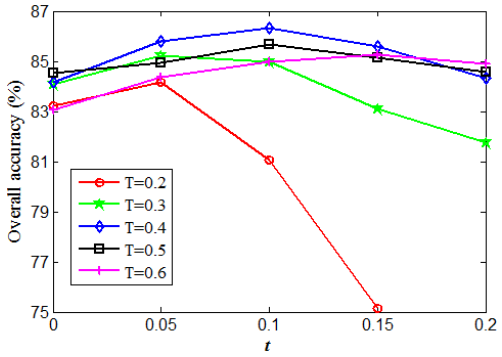


Fig. 10. Overall accuracy versus margin thresholds over validation set. T and t are selected from among $\{0.2, 0.3, 0.4, 0.5, 0.6\}$ and $\{0.00, 0.05, 0.10, 0.15, 0.20\}$, respectively, excluding $T = t = 0.2$. It turns out that the classification performance peaks when $T = 0.4, t = 0.1$.

D. Network Architecture and Parameter Selection

GoogLeNet [75], which is lightweight and usually performs well, is selected as the branches of EM-CNN, with the 1024-dimensional features $f(x)$ and $f(x')$ "living" in the last pooling layers.

In similarity distribution learning network, the FC layer and output layer has 256 and 30 neurons, respectively. In other words, we set $L = 30$. This is based on our previous study [1], which dealt with similarity learning through deep ordinal classification, reaching the conclusion that retrieval performance is not sensitive to the number of subintervals greater than 15 and a value around 30 is appropriate.

Furthermore, Wen *et al.* [49] found that classification performance remains largely stable when the learning rate of class centers varies from 0.01 to 1.0. So, we set $R = r = 0.2$. Besides, the small positive number ϵ in (46) and (52) takes the value 0.0001.

As for the margin thresholds, we determine T and t over the NR validation set, and the empirically optimal values are then used for all our experiments. $T \in \{0.2, 0.3, 0.4, 0.5, 0.6\}$ and $t \in \{0.00, 0.05, 0.10, 0.15, 0.20\}$ are tried, excluding $T = t = 0.2$. Over the NR training set, we train GoogLeNet with MLPNC loss (i.e., model 8 in Section V, which depends greatly on T and t, is used), and summarize the validation results in Fig. 10 (note that in this article, the overall accuracy, which is the ratio of the number of correctly classified samples to the total number of samples, acts as the evaluation measure of ASC). It turns out that the classification performance peaks when $T = 0.4, t = 0.1$, so we set T and t to 0.4 and 0.1, respectively. Besides, it is found that when T takes a small value such as 0.2, classification performance is sensitive to the values of t. This is easily explained—T and t are interclass and intraclass margins, respectively, so it is necessary that there is a reasonable difference between them.

Loss weights λ_1 and λ_2 are determined over the NR validation set, too. We first fix λ_2 to 4.0, and vary λ_1 among $\{0, 0.025, 0.05, 0.1, 0.2, 0.4, 0.6, 0.8, 1.0\}$, then we fix λ_1 to 0.2, and vary λ_2 from 0 to 8. The validation results are shown in Fig. 11. Apparently, the model performs worst when $\lambda_1 = 0$ or $\lambda_2 = 0$, indicating that both MLPNC loss and

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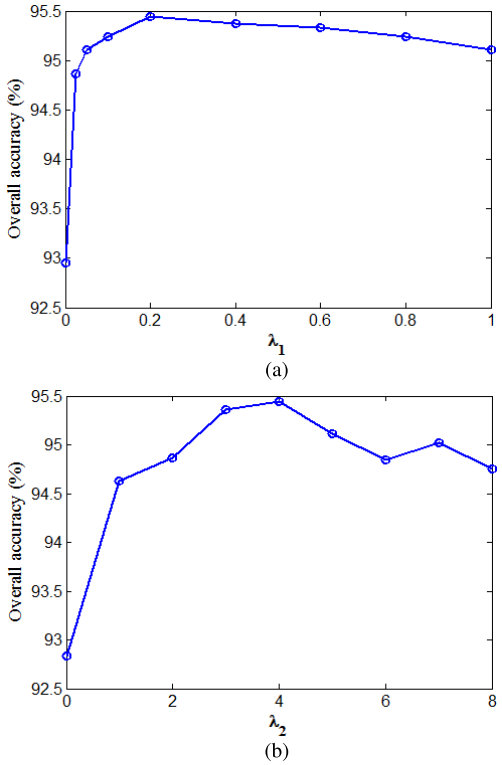


Fig. 11. Overall accuracy versus loss weights over the validation set. (a) λ_2 is fixed to 4.0, and λ_1 is varied among $\{0, 0.025, 0.05, 0.1, 0.2, 0.4, 0.6, 0.8, 1.0\}$. The optimal value of λ_1 is 0.2. (b) λ_1 is fixed to 0.2, and λ_2 is varied from 0 to 8. The optimal value of λ_2 is 4.

WDBSD loss contribute to performance improvement. It also turns out that after a sharp increase, overall accuracy is insensitive to λ_1 greater than 0.025 and λ_2 greater than 1.0, and $\lambda_1 = 0.2$ and $\lambda_2 = 4.0$ seem the best choices. So we set $\lambda_1 = 0.2$ and $\lambda_2 = 4.0$.

Moreover, SGD, as mentioned above, acts as our optimization algorithm, with a batch size of 64 (in image pairs), a weight decay of 0.0005, and a momentum of 0.9. Our learning rate ρ starts with 0.001, then it is divided by 10 every 10 epochs.

IV. Experimental Results of Aerial Image Retrieval

This section presents and analyzes the experimental results of CBAIR.

A. Evaluation Measures

We use six measures to evaluate retrieval performance: average normalized modified retrieval rank (ANMRR), mean average precision (mAP), precision at cutoff k (P@k), the interpolated precision–recall (P–R) curve, Normalized Discounted Cumulative Gain at cutoff k (NDCG@k) [76], and satisfaction score. The first four measures concern coarse-level similarity, their detailed definitions can be found in [1]; whereas NDCG@k is specific to retrieval evaluation involving multilevel similarity and is defined as

$$\text{NDCG@}k = \frac{1}{\Upsilon}\sum_{i=1}^{k}\frac{2^{\upsilon_i}-1}{\ln(1+i)} \tag{53}$$

where k is the truncated position in a ranking list, Υ is a normalization constant to guarantee that NDCG@k for the correct ranking equals 1, and υ_i is the similarity level roughly indicating the similarity between a query and the ith retrieved image in the ranking list. If these two images belong to the same subclass, $\upsilon_i = 2$; if they only belong to the same class, $\upsilon_i = 1$; otherwise, $\upsilon_i = 0$.

The first five kinds of evaluations are performed by using each test image to query against the rest of the test images and reporting the average.

The sixth measure, that is, satisfaction score, is designed by ourselves, and the satisfaction scores are provided by users. After all, users have the last word on whether or not retrieval performance is satisfying, especially when our proposed fine-grained retrieval model is concerned, because the concept of "fine-grained retrieval through subclasses" is given by us, and there has not been a unified view on what subclasses are—researchers might not even agree that subclasses exist.

We first set five satisfaction levels, with levels one to five meaning "unsatisfactory," "satisfactory," "good," "very good," and "excellent," respectively, and then we ask 200 people to score their satisfaction. More specifically, each person randomly selects five images from each test set as queries, and grades the corresponding retrieval results after the system provide a ranking list of length 10. Finally, 1000 scores ($200 \times 5 = 1000$) over each data set are averaged, yielding the final satisfaction score that is reported in this section.

B. Clustering Results

The CHC-based clustering algorithm delineated in Section II-A creates subclasses for each class, Fig. 12 shows some clustering results: the training images from the "shipping yard" class of PatternNet are split into four subclasses [Fig. 12(a)], and those from the "harbor" class of UCM are split into three subclasses [Fig. 12(b)].

We can see that the images assigned to one subclass are quite similar in appearance, color, and background, and clustering results are roughly consistent with human perception. Besides, it is apparent that large variations exist across subclasses. We also try DBC and SC. Although different clustering evaluation criteria may lead to slightly different numbers of subclasses, similar findings can be reached.

In order to investigate how CHC, DBC, and SC impact retrieval, we equip GoogLeNet with MLPNC loss that is closely related to within-class clustering (i.e., model 8 in Section IV-C is used here), and summarize the experimental results in Tables III and IV. It turns out that 1) the effect of clustering evaluation criteria on retrieval performance is not significant and 2) on the whole, CHC holds a slim performance advantage over DBC and SC.

In light of these results, we use only CHC in other experiments.

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TABLE III

EFFECT OF CLUSTERING EVALUATION CRITERIA ON RETRIEVAL PERFORMANCE (OVER PATTERNNET). A SMALLER VALUE OF ANMRR INDICATES A BETTER PERFORMANCE, AND THE OPPOSITE IS TRUE FOR MAP (%), P@k (%), AND NDCG@k (%)

Criterion	ANMRR	mAP	P@5	P@10	P@50	P@100	P@1000	NDCG@320
CHC	**0.0629**	**92.11**	**97.32**	96.91	**95.51**	**93.56**	**15.72**	**93.18**
DBC	0.0647	91.71	97.29	96.90	95.31	**93.56**	15.70	92.66
SC	0.0634	91.85	97.30	**96.93**	95.35	93.54	15.71	91.43

TABLE IV

EFFECT OF CLUSTERING EVALUATION CRITERIA ON RETRIEVAL PERFORMANCE (OVER UCM). A SMALLER VALUE OF ANMRR INDICATES A BETTER PERFORMANCE, AND THE OPPOSITE IS TRUE FOR MAP (%), P@k (%), AND NDCG@k (%)

Criterion	ANMRR	mAP	P@5	P@10	P@50	P@100	P@1000	NDCG@100
CHC	**0.1511**	**81.24**	**90.44**	**88.42**	**76.08**	**43.61**	**4.90**	88.57
DBC	0.1532	81.18	90.38	88.35	76.03	43.54	4.90	**88.62**
SC	0.1558	80.83	90.24	88.22	75.65	43.51	4.90	88.09

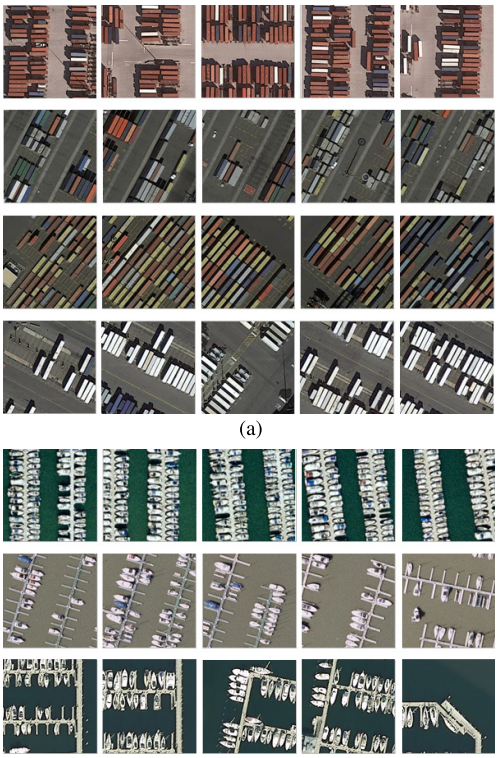


Fig. 12. Some clustering results. Images in the same subclass are similar in appearance, color, and background, and clustering results are roughly consistent with human perception. (a) "Shipping yard" images are clustered into four subclasses. (b) "Harbor" images are clustered into three subclasses.

C. Performance Comparison Among Different Models

This section makes a comparison among the following 11 models: models 1, 2, 3, 4, 5, and 8 are GoogLeNet coupled with CE loss, "CE + center" loss, "CE + triplet" loss, "CE + L1PNC" loss, "CE + MLPNC" loss, and MLPNC loss, respectively; models 6 and 7 are EM-CNN, conducting retrieval on the basis of 1024-dimensional FVs and predicted SSs, respectively (we use "FV" and "SS" to distinguish between these two models); models 9–11 have exactly the same network architecture as EM-CNN, but they are coupled with WDBSD loss, "CE + WDBSD" loss, and "MLPNC + WDBSD" loss, respectively, and they all conduct retrieval based on the predicted SSs.

Note that the training procedure of model 3 is basically the same as that in [26], except that we replace contrastive embedding with triplet embedding. Besides, the center loss should be bound to CE loss; otherwise, the learned deep features and centers will degrade to zeros [49]. For a fair comparison, some of the 11 models are also equipped with a "bound-together" loss. As a matter of fact, except center loss, all losses involved can be used alone.

Retrieval performance in terms of ANMRR, mAP, P@k (we let $k = 5, 10, 50, 100, 1000$, as most researchers did), and NDCG@k (we let k equal twice the number of test images of each class; note that in CBAIR, NDCG@k has never been used before) is summarized in Tables V and VI. Besides, Table VII gives the average satisfaction scores. Furthermore, Figs. 13 and 14 present the P–R curves, and Fig. 15 shows some images on the top-5 ranking lists.

Experimental results reveal that model 1 $<$ model 2 $\approx$ model 3 $<$ model 4 $<$ model 5 $<$ model 6 $<$ model 7 ("$<$" and "$\approx$" mean "has a weaker performance than" and "approximates," respectively). In other words, center loss $\approx$ triplet loss $<$ L1PNC loss $<$ MLPNC loss $<$ "MLPNC + WDBSD" loss. On the basis of these observations, we reach the following conclusions.

1) Developed from center learning and triplet embedding, center-metric learning outperforms them both not only in terms of traditional evaluation measures but also in terms of users' satisfaction. Moreover, Fig. 15 demonstrates that positive–negative center loss makes different classes far apart, thereby overcoming center loss' drawbacks—model 4 equipped with L1PNC loss excludes other classes' images from the ranking lists; in contrast, "alien" images occur in model 2's top-5 lists. More than that, our positive–negative center loss neatly sidesteps the troubles going with triplet creation.

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TABLE V

RETRIEVAL PERFORMANCE COMPARISON AMONG DIFFERENT MODELS (OVER PATTERNNET). A SMALLER VALUE OF ANMRR INDICATES A BETTER PERFORMANCE, AND THE OPPOSITE IS TRUE FOR MAP (%), P@k (%), AND NDCG@k (%)

Evaluation Measure	Model 1 CE	Model 2 CE+Center	Model 3 CE+Triplet	Model 4 CE+L1PNC	Model 5 CE+MLPNC	Model 6 C-M-W (FV)	Model 7 C-M-W (SS)	Model 8 MLPNC	Model 9 WDBSD	Model 10 CE+WDBSD	Model 11 MLPNC+ WDBSD
ANMMR	0.0902	0.0716	0.0719	0.0523	0.0261	0.0119	**0.0021**	0.0629	0.0457	0.0366	0.0215
mAP	88.47	90.80	90.79	93.25	96.44	98.34	**99.51**	92.11	94.06	95.45	97.04
P@5	96.73	97.09	97.11	97.65	98.41	98.93	**99.71**	97.32	97.96	98.14	98.82
P@10	96.31	96.59	96.66	97.25	98.19	98.81	**99.62**	96.91	97.66	97.85	98.76
P@50	93.99	94.91	94.93	95.91	97.48	98.59	**99.60**	95.51	96.49	96.88	98.34
P@100	91.32	92.86	92.79	94.63	96.77	98.38	**99.57**	93.56	95.31	96.29	97.78
P@1000	15.63	15.69	15.69	15.74	15.85	15.90	**15.90**	15.72	15.77	15.82	15.86
NDCG@320	89.12	90.36	91.17	92.38	94.76	95.98	**96.34**	93.18	92.62	93.86	95.01

TABLE VI

RETRIEVAL PERFORMANCE COMPARISON AMONG DIFFERENT MODELS (OVER UCM). A SMALLER VALUE OF ANMRR INDICATES A BETTER PERFORMANCE, AND THE OPPOSITE IS TRUE FOR MAP (%), P@k (%), AND NDCG@k (%)

Evaluation Measure	Model 1 CE	Model 2 CE+Center	Model 3 CE+Triplet	Model 4 CE+L1PNC	Model 5 CE+MLPNC	Model 6 C-M-W (FV)	Model 7 C-M-W (SS)	Model 8 MLPNC	Model 9 WDBSD	Model 10 CE+WDBSD	Model 11 MLPNC+ WDBSD
ANMMR	0.1901	0.1479	0.1196	0.1029	0.0605	0.0349	**0.0203**	0.1511	0.0694	0.0522	0.0412
mAP	75.59	82.03	84.95	87.04	91.71	94.78	**97.01**	81.24	90.74	92.74	94.03
P@5	88.13	90.72	92.47	93.42	95.91	97.36	**97.84**	90.44	95.42	96.39	97.01
P@10	85.54	88.84	90.96	92.06	95.02	96.81	**97.65**	88.42	94.48	95.65	96.40
P@50	69.85	76.98	80.11	82.04	86.65	90.45	**93.31**	76.08	85.72	87.92	89.53
P@100	42.74	43.69	44.64	45.42	47.26	48.25	**48.69**	43.61	46.88	47.58	48.01
P@1000	4.90	4.90	4.90	4.90	4.90	4.90	**4.90**	4.90	4.90	4.90	4.90
NDCG@100	83.16	86.77	88.56	90.61	91.99	93.67	**94.08**	88.57	91.68	92.44	93.57

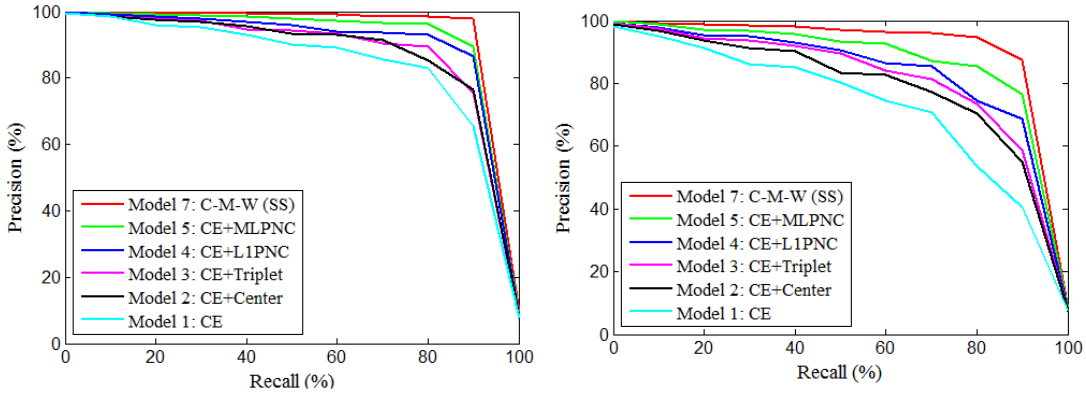


Fig. 13. Interpolated P–R curves for comparison of different losses. It turns out that center loss ≈ triplet loss < L1PNC loss < MLPNC loss < "MLPNC + WDBSD" loss ("<" and "≈" mean "has a weaker performance than" and "approximates," respectively; a curve farther away from the origin indicates a better performance). (a) Over PatternNet. (b) Over UCM.

2) MLPNC loss, which integrates fine-grained L2PNC loss with coarse-grained L1PNC loss, is able to recognize within-class variations. This can be intuitively validated by Fig. 15—compared to L1PNC loss, MLPNC loss includes fewer images of other subclasses in its ranking lists. Forced to find subtle differences among images, model 5 has better generalization ability than model 4 that is only equipped with class-level L1PNC loss (see Tables V–VII, and Fig. 13).
3) WDBSD loss contributes to EM-CNN's fine discrimination ability—in Fig. 15, all the images retrieved by model 7 come from the same subclasses as the queries. This can be attributed to WDBSD loss' fine-grained similarity predictions and modeling of similarity uncertainty.
4) From the finding that model 7 outperforms model 6 (see Tables V and VI), we conclude that SS prediction, which is retrieval-oriented, provides more valuable information for CBAIR than deep features.

Furthermore, we divide into three groups the models related to CE loss, MLPNC loss, and WDBSD loss: group 1 includes models 1, 8, and 9, which are all single-task CNNs; group 2 includes models 5, 10, and 11, which are all two-task CNNs; group 3 includes models 6 and 7, which are both three-task CNNs. The results in Tables V and VI and Fig. 14 indicate that group 1 < group 2 < group 3, that is to say, two-task models

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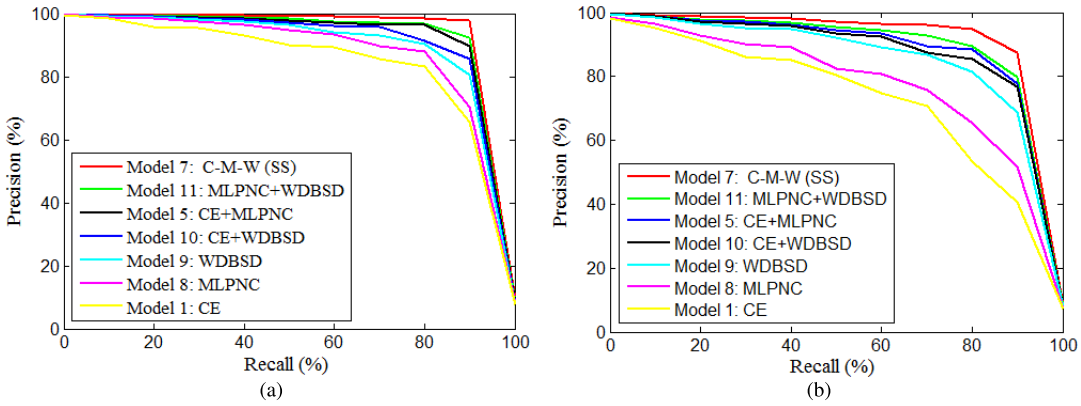


Fig. 14. Interpolated P–R curves for comparison of single-task models and multitask models. It turns out that two-task models perform better than single-task models, and three-task models perform best (a curve farther away from the origin indicates a better performance). (a) Over PatternNet. (b) Over UCM.



Fig. 15. Some retrieval results in the form of top-5 ranking lists. "T" ("true"), "U" ("unsatisfactory"), and "F" ("false") mean that the retrieved image and the query belong to the same subclass, different subclasses (and the same class), and different classes, respectively. From top to bottom, the ranking lists correspond to models 1, 2, 3, 4, 5, and 7, respectively. "F" does not occur in the ranking lists of model 3 or 4, indicating that triplet loss and L1PNC loss successfully increase between-class distances, this is in contrast to models 1 and 2. "U" occurs less frequently in model 5 than in model 4, indicating that MLPNC loss endows CNNs with fine discrimination ability. Model 7 excludes "U" from its top-5 ranking lists, indicating that WDBSD loss further enhances CNNs' fine discrimination ability. (a) "Swimming pool" query from PatterNet and its top-5 ranking lists. (b) "Dense residential" query from UCM and its top-5 ranking lists.

perform better than single-task models, and three-task models perform best. This proves that multitask learning is useful and effective, explaining, to some extent, why our EM-CNN has competitive performance.

In addition, it turns out that when used separately, both MLPNC loss and WDBSD loss are much superior to the most widely used CE loss; in particular, the stand-alone WDBSD loss has an absolute performance advantage over the "bound-together" losses associated with models 2–4. This indicates that our proposed MLPNC loss and WDBSD loss are promising.

D. Results of Similarity Distribution Learning

Fig. 16 shows three image pairs, their ground truth and predicted similarity distributions. We can see that the predicted similarity distributions roughly have the same shapes as ground truth similarity distributions.

Moreover, we compare similarity distribution learning with another two similarity learning methods: deep regression [52], [53] and ordinal classification [1], [77]. These two approaches adopt the same network architecture as EM-CNN, except that the last two layers in the similarity distribution learning network are adjusted. In deep regression, the average SSs

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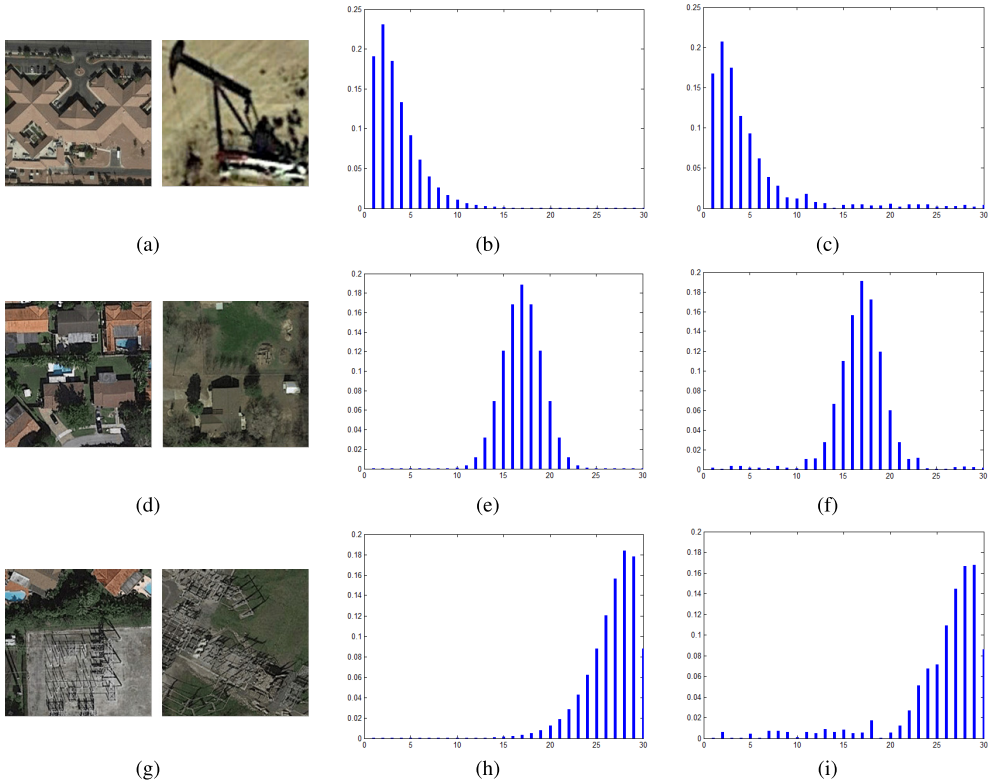


Fig. 16. Some image pairs and their ground truth and predicted similarity distributions. The predicted similarity distributions have roughly the same shape as ground truth similarity distributions. (a) First image pair. (b) First image pair's ground truth similarity distribution (Gamma distribution). (c) First image pair's predicted similarity distribution. (d) Second image pair. (e) Second image pair's ground truth similarity distribution (Gaussian distribution). (f) Second image pair's predicted similarity distribution. (g) Third image pair. (h) Third image pair's ground truth similarity distribution (vertically flipped Gamma distribution). (i) Third image pair's predicted similarity distribution.

TABLE VII
AVERAGE SATISFACTION SCORES. LARGER VALUES MEAN HIGHER SATISFACTION

Dataset	Model 1 CE	Model 2 CE+Center	Model 3 CE+Triplet	Model 4 CE+L1PNC	Model 5 CE+MLPNC	Model 7 C-M-W (SS)
PatternNet	3.42	3.87	4.08	4.35	4.56	**4.86**
UCM	3.17	3.49	3.85	4.11	4.42	**4.75**

given by (23) serve as ground truths, so correspondingly, the prediction layer has only one neuron; and regression loss formulated as the sum of squared distances between ground truths and predicted values replaces WDBSD loss. In ordinal classification, the prediction layer has 30 neurons responsible for outputting similarity pseudolabels, and weighted Wasserstein ordinal loss [1] replaces WDBSD loss.

In the 11 models mentioned in Section IV-C, there are five models related to similarity distribution learning. We try three different similarity learning approaches for them, and report the experimental results in Fig. 17. Our observations and analyses are listed as follows.

1) Over the two data sets and in the five models, our proposed similarity distribution learning invariably outperforms deep regression and ordinal classification, both of which completely ignore uncertainty with regard to similarity between two images. This indicates that modeling uncertainty is reasonable and helps improve retrieval performance.
2) On the whole, ordinal classification is superior to deep regression. This is consistent with other researchers' finding that deep regression is sensitive to outliers, and thereby causing unstable predictions [1], [77].

E. Effect of the Number of Experts on Performance

In this section, we investigate how the number of CNN experts influences retrieval performance. We let M_1 vary among $\{4, 6, 8, 10, 12\}$ (when $M_1 = i$, the first ith experts in Table II are selected), and present the results in Fig. 18. Our observations are as follows.

1) The larger the number of experts, the better the retrieval performance. This is not surprising—different experts complement one another, more experts provide more powerful information and, hence, the resulting estimations of similarity distributions are more reliable.
2) The number of experts has more effect on single-task models than on multitask ones. A possible explanation is that in single-task CNNs, training is completely controlled by one task, that is, similarity distribution learning, so retrieval performance is largely determined by experts; in contrast, in a multitask scenario where multiple tasks jointly supervise optimization process, experts play a less important role in training, therefore,

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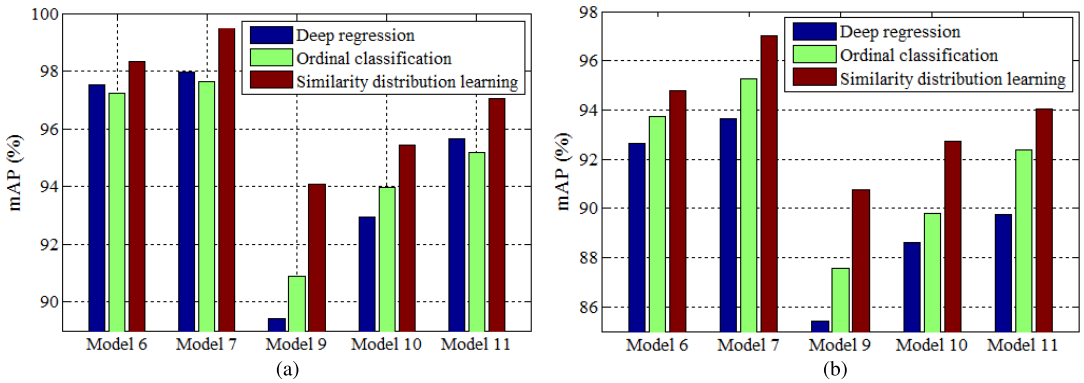


Fig. 17. Retrieval Performance comparison among different methods of similarity learning. Similarity distribution learning outperforms both deep regression and ordinal classification in terms of retrieval Performance. (a) Over PatternNet. (b) Over UCM.

TABLE VIII

COMPARISON WITH STATE-OF-THE-ART METHODS (OVER PATTERNNET). A SMALLER VALUE OF ANMRR INDICATES A BETTER PERFORMANCE

Method	ANMRR	# Network Parameters	Average Retrieval Time (s)	Training Set Size
UFL [36]	0.6584	-	-	**0%**
Gabor texture [36]	0.6439	-	-	**0%**
VLAD [36]	0.5677	-	-	**0%**
VGG-F [36]	0.2995	60 M	0.4318	**0%**
GoogLeNet-m [40]	0.2784	**7 M**	**-**	**0%**
ResNet50 [36]	0.2584	26 M	0.2020	**0%**
LDCNN [34, 36]	0.2416	37 M	-	80%
SBS-CNN [1]	0.2185	10M	**0.0667**	70%
DML [44]	0.0030	26 M	-	80%
EM-CNN [this work]	**0.0021**	**7 M**	0.0783	20%

TABLE IX

COMPARISON WITH STATE-OF-THE-ART METHODS (OVER UCM). A SMALLER VALUE OF ANMRR INDICATES A BETTER PERFORMANCE

Method	ANMRR	# Network Parameters	Average Retrieval Time (s)	Training Set Size
ResNet101 [36]	0.356	45 M	0.0411	**0%**
Fine-tuned VGG-M [34]	0.329	100 M	0.0761	**0%** [1]
Feature combinations [40]	0.2915	245 M	-	**0%**
GoogLeNet+PCA [37]	0.285	**7 M**	**-**	**0%** [1]
SBS-CNN [1]	0.2683	10 M	**0.0127**	**0%** [2]
SatResNet-50 [35]	0.239	26 M	0.0397	**0%** [1]
Discriminator NW [78]	0.09	64 M	-	80%
Weighted distance [38]	0.0404	26 M	0.0397	80%
DML [44]	0.0223	26 M	-	50%
EM-CNN [this work]	**0.0203**	**7 M**	0.0165	50%

[1] Fine-tuned over AID.
[2] Fine-tuned over NR.

the number of experts makes a smaller impact on performance.

Moreover, as mentioned in Section III-B, we fix the value of M_2 at 3, and select 3 experts to create subclasses for fine-grained retrieval and classification. We try utilizing different experts (but always with one expert per CNN family) and find that the resulting subclasses differ very slightly and have little effect on retrieval performance. Therefore, we always use the three experts: CaffeRef, VGG-VD-16, and ResNet-50.

F. Comparison With State-of-the-Art

In this section, we compare EM-CNN with state-of-the-art methods. The comparison is based on retrieval performance (ANMRR), network size, average retrieval time, and training set size, and hence is comprehensive. For the sake of fairness, the average retrieval times are all computed under the same hardware configuration and on the same development platform—we ourselves implement the algorithms whose source codes are publicly available and report the time required to perform retrieval.

Tables VIII and IX summarize the results, from which we can see that EM-CNN outperforms all existing approaches not only in terms of retrieval performance, but also in terms of network size (it should be stressed that although we use many CNN experts, the resulting model is small—experts squeeze their powerful information into EM-CNN and "get away" after computation of similarity distributions is finished). Besides, our average retrieval time ranks second over both data sets, slightly slower by 0.0116 and 0.0038 s than the fastest models over PatternNet and UCM, respectively. Compared to the methods that have a competitive performance, our training set sizes are the smallest.

In short, the thorough comparison reveals that our EM-CNN has distinct advantages over the state-of-the-art methods.

Finally, it should be noted that the traditional non-CNN methods are not listed in Table VIII or IX, because they are usually markedly inferior to CNNs and comparing them does not make much sense.

V. EXPERIMENTAL RESULTS OF ASC

Experimental results of ASC are presented and analyzed in this section.

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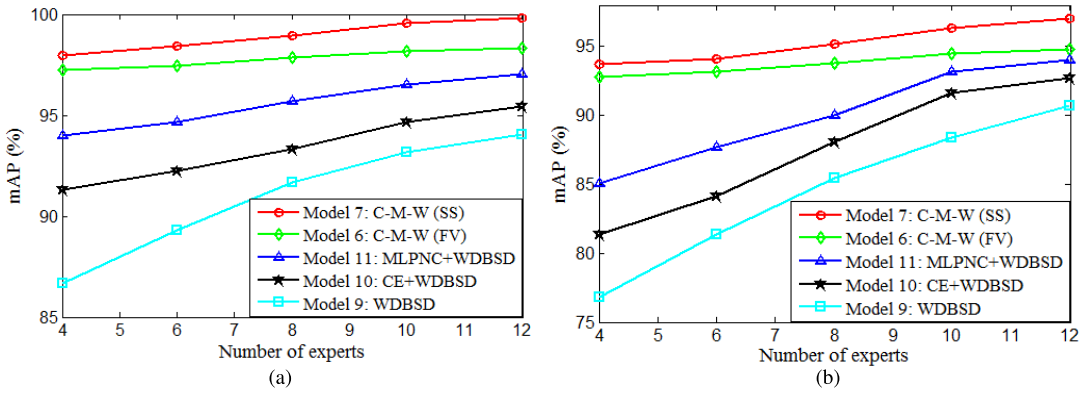


Fig. 18. Retrieval Performance versus number of experts. Retrieval performance increases with the number of experts. (a) Over PatternNet. (b) Over UCM.

TABLE X
CLASSIFICATION PERFORMANCE COMPARISON AMONG DIFFERENT MODELS

Dataset	Model 1 CE	Model 2 CE+Center	Model 3 CE+Triplet	Model 4 CE+L1PNC	Model 5 CE+MLPNC	Model 6 C-M-W	Model 8 MLPNC	Model 9 WDBSD	Model 10 CE+WDBSD	Model 11 MLPNC+ WDBSD
NR	85.03±0.20	88.66±0.19	89.42±0.18	91.56±0.20	92.79±0.17	**95.17±0.17**	86.18±0.22	88.33±0.23	93.16±0.24	94.27±0.22
AID	91.48±0.33	93.28±0.31	93.23±0.26	94.35±0.26	95.24±0.28	**97.21 ±0.27**	90.24±0.29	92.68±0.33	96.15±0.32	95.89±0.30

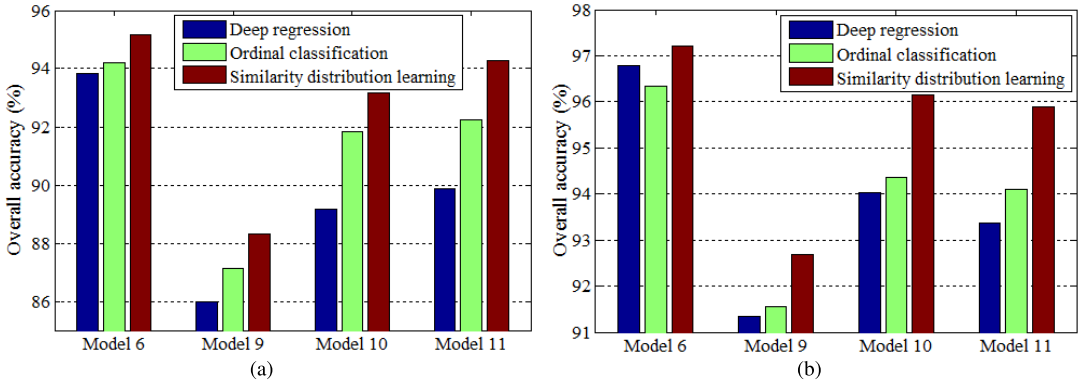


Fig. 19. Classification performance comparison among different methods of similarity learning. Similarity distribution learning outperforms both deep regression and ordinal classification in terms of classification Performance. (a) Over NR. (b) Over AID.

Over each data set, we perform ten rounds of experiments, each round with a different data set division. We report classification performance as the mean and standard deviation of overall accuracies over ten rounds.

Model 7 is not applicable to ASC, so it is not considered in this section. Besides, since model 8, 9, or 11 cannot yield classification results in an end-to-end manner, for the purpose of a fair comparison, we always follow the same classification procedure: extract features from the trained model, and then use linear support vector machines for classification.

Performance comparison among ten models is carried out, and the results are shown in Table X. Broadly speaking, these results conform to our findings and conclusions given in Section IV-C: L1PNC loss that concentrates on both within-class compactness and between-class large-distance performs better than center loss and triplet loss; MLPNC loss that pays attention to within-class variations performs better than L1PNC loss; and "MLPNC + WDBSD" loss that takes fine-grained similarity into consideration performs better than MLPNC loss; besides, two-task CNNs prevail over single-task CNNs, whereas EM-CNN, which is a three-task model, prevails over two-task CNNs.

In addition, comparison among three methods of similarity learning is made, and Fig. 19 presents the results. It turns out that, as in CBAIR, our proposed similarity distribution learning surpasses another two approaches, that is, deep regression and ordinal classification.

Furthermore, a comparison with existing methods is made. For the sake of comprehensiveness, we summarize many models proposed in recent years, see Tables XI and XII. The results show that our EM-CNN outperforms all state-of-the-art approaches in terms of classification performance.

VI. DISCUSSION

In this section, we discuss several problems involved in our proposed methods and give some possibly fruitful directions for future research.

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TABLE XI
COMPARISON WITH STATE-OF-THE-ART METHODS (OVER NR)

Method	Overall Accuracy (%)
Two-stream deep fusion [79]	83.16±0.18
Bag of convolutional features [19]	84.32±0.17
Generative framework [28]	88.6±0.31
VGG-VD16+MSCP [14]	88.93±0.14
Fine-tuned VGGNet-16 [11]	90.36±0.18
Discriminative CNN [26]	91.89±0.22
Siamese ResNet50 [80]	92.28±3.78
Triplet networks [2]	92.33±0.20
DDRL-AM [81]	92.46±0.09
SF-CNN with VGGNet [23]	92.55±0.14
Inception-v3-CapsNet [82]	92.6±0.11
ConvNet+Context Aggregation [83]	93.47±0.26
ADSSM [15]	94.29±0.14
HW-CNN [29]	94.38±0.16
An ensemble of CNNs [84]	94.51±0.21
EM-CNN [this work]	**95.17±0.17**

TABLE XII
COMPARISON WITH STATE-OF-THE-ART METHODS (OVER AID)

Method	Overall Accuracy (%)
Transferred VGGNet-16 [3]	89.64±0.36
salM^{3}LBP-CLM [85]	89.76±0.45
Generative framework [28]	91.35±0.83
Feature fusion by addition [22]	91.87±0.36
Two-stream deep fusion [79]	92.32±0.41
ARCNet-VGG16 [24]	93.10±0.55
MSP+Fisher vector [86]	93.9
VGG-VD16+MSCP [14]	94.42±0.17
Two-stage deep feature fusion [25]	94.65±0.33
ResNet+WSPM-CRC [87]	95.11
Rearranged local features [12]	95.29±1.28
Binary patterns encoded CNNs [18]	95.73±0.16
Two-stream fusion [17]	95.99±0.35
Inception-v3-CapsNet [82]	96.32±0.12
SF-CNN with VGGNet [23]	96.66±0.11
Discriminative CNN [26]	96.89±0.10
HW-CNN [29]	96.98±0.33
Inception-DenseNet [88]	97.15
EM-CNN [this work]	**97.21±0.27**

Our EM-CNN is closely associated with CNN experts—both subclass creation and similarity distribution generation depend on experts that have been pretrained and fine-tuned with CE loss. However, CE loss is deemed to ignore intraclass compactness, a weakness we aim to compensate for in this article. So, why can experts equipped with the defective CE loss provide the basis for optimizing loss functions that we hope are free of CE loss' weakness? Multiple-CNN information fusion lies behind the seemingly impossible achievement: various CNNs provide complementary information, which are consolidated by us to give a more discriminative representation and hence better capture the characteristics of aerial images; the combined deep features are so powerful that they can help to produce a new model that gets rid of CE loss' faults. This has been justified by our previous study [29], which demonstrates that combining multiple pretrained CNNs is necessary and effective. In fact, our past success in overcoming CE loss' shortcomings with the help of several CNNs pretrained with CE loss [29] has encouraged us to follow a similar route in this article.

Empirically, we have verified that multitask learning does enhance the generalization ability of the model. However, we do not know if the enhancement results from intertask relationship—as stated in Section I-C, we cannot even confirm that relatedness among tasks exists. Since task relatedness is an essential precondition for positive multitask learning, it is important to find effective ways of explicitly modeling inherent relationship among tasks. Long *et al.* [89] made such a preliminary attempt by placing a tensor norm distribution as a prior on the parameter tensor in the FC layers. This is quite enlightening, and in the near future, we will investigate their proposed multilinear relationship networks, with the hope that EM-CNN can be improved to automatically learn quantitative task relatedness from training data.

Furthermore, we have determined task weights λ_1 and λ_2 via brute-force search. This is time-consuming and cannot guarantee that the selected values are optimal. In the near future, we will try the dynamic-weighting scheme [57], which enables CNNs to learn to dynamically assign loss weights to all tasks during training.

Besides, our subclasses are prepared beforehand and remain unchanged throughout the training process. To dynamically update subclasses using the just-learned deep features may further enhance EM-CNN. This can form the focus of our future research.

Finally, it is desirable that clustering schemes should take human prior knowledge into account. For example, airplanes of the same type will be grouped together; circular, oval, and rectangular swimming pools will belong to different subclasses; and so on. In this way, human beings participate in defining what is "fine-grained" and, hence, the resulting clustering results will make more sense. Therefore, it is of interest to investigate how human opinions can be embedded into clustering algorithms.

VII. CONCLUSION

In recent years, CNNs have played a predominant role in CBAIR and ASC because of their markedly superior performance.

However, CNN-based methods for CBAIR and/or ASC ignore intraclass variations that are usually very large, and thereby overlooking the possibility of fine retrieval and classification. To address this issue, we propose a novel metric learning method referred to as center-metric learning, and couple it with a new kind of loss referred to as positive–negative center loss, which is composed of L1PNC loss, L2PNC loss, and MLPNC loss. Positive–negative center loss has the following advantages: first, it can not only reduce within-class scatterness

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but also increase between-class distance, remedying center loss' deficiencies; second, it avoids the trouble of producing a tremendous number of image pairs/triplets and mining hard samples, overcoming triplet loss and contrastive loss' shortcomings; third, L2PNC loss and MLPNC loss can cope with intraclass variations by virtue of several experts, compelling CNNs to make fine visual discriminations.

Besides, existing CBAIR models consider the similarity between two images as a fixed value, unaware that similarity is by its very nature uncertain. To solve this problem, for the first time, we propose similarity distributing learning, which, with the help of some CNN experts, elegantly embeds uncertainty into the training process. Similarity distributing learning provides fine-grained similarity predictions, further enhancing CNNs' ability to handle intraclass variations.

Furthermore, three tasks, that is, center-metric learning, similarity distributing learning, and ASC are incorporated into one CNN, thereby benefitting one another and leading to a better generalization capability. The resulting model, EM-CNN, can discriminate subtle differences among aerial images belonging to the same class, so we say it has as good eyesight as an eagle.

We conduct extensive experiments over four publicly available aerial image sets, including PatternNet, UCM, NR, and AID, and it turns out that our EM-CNN outperforms all existing methods in terms of performance.

ACKNOWLEDGMENT

Part of this work was carried out while the first author was visiting the Center for Pattern Recognition and Machine Intelligence (CENPARMI), Concordia University, Canada. She wishes to express her gratitude to CENPARMI for its warm hospitality. And all the authors would like to thank the editors and anonymous reviewers for their valuable and constructive comments.

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